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United States Patent	12418756
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lovchinsky; Igor et al.

System and method for enhancing speech of target speaker from audio signal in an ear-worn device using voice signatures

Abstract

An ear-worn device is provided that operates to isolate and individually treat the received speech of a target speaker or multiple target speakers from an audio input signal detected in a multi-speaker environment. The ear-worn device uses a machine learning model that receives a voice signature of each of one or more target speakers as input signals, to identify and isolate the component of the audio input signal attributable to the target speaker(s). Once isolated, the target speaker's speech may be enhanced, de-emphasized, or otherwise processed in a manner desired by the wearer of the ear-worn device. The wearer may use an external electronic device, e.g., a phone, to select one or more target speakers in a conversation and/or configure various settings associated with processing the speech on the ear-worn device.

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Appl. No.: 18/097154

Filed: January 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230306982 A1	Sep. 28, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17576718 20220114 US 11950056 child-doc US 18097154
continuation-in-part parent-doc US 17576746 20220114 US 12075215 child-doc US 18097154
continuation-in-part parent-doc US 17576893 20220114 US 11818547 child-doc US 18097154
continuation-in-part parent-doc US 17576899 20220114 US 11832061 child-doc US 18097154
continuation-in-part parent-doc WO PCT/US2022/012567 20220114 PENDING child-doc US
18097154
us-provisional-application US 63305676 20220201

Publication Classification

Int. Cl.: **H04R25/00** (20060101); **G06F3/16** (20060101); **G10L17/18** (20130101); **G10L21/0272**
(20130101); **G10L21/028** (20130101); **G10L21/0316** (20130101); **G10L21/0364**
(20130101); **G10L25/78** (20130101)

U.S. Cl.:

CPC **H04R25/507** (20130101); **G06F3/165** (20130101); **G10L17/18** (20130101);
G10L21/0272 (20130101); **G10L21/028** (20130101); **G10L21/0316** (20130101);
G10L21/0364 (20130101); **G10L25/78** (20130101); **H04R25/558** (20130101);
H04R25/604 (20130101); **H04R25/609** (20190501); H04R2225/43 (20130101);
H04R2225/55 (20130101); H04R2430/01 (20130101)

Field of Classification Search

CPC: H04R (25/507); H04R (25/609); H04R (25/558); H04R (25/604); H04R (2225/43); H04R
(2225/55); H04R (2430/01)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part, claiming the benefit under 35 U.S.C. § 120, of U.S. application Ser. No. 17/576,718, entitled “Method, Apparatus and System for Neural Network Hearing Aid,” filed Jan. 14, 2022, which is herein incorporated by reference in its entirety. (2) This application is a continuation-in-part, claiming the benefit under 35 U.S.C. § 120, of U.S. application Ser. No. 17/576,746, entitled “Method, Apparatus and System for Neural Network Hearing Aid,” filed Jan. 14, 2022, which is herein incorporated by reference in its entirety. (3) This application is a continuation-in-part, claiming the benefit under 35 U.S.C. § 120, of U.S. application Ser. No. 17/576,893, entitled “Method, Apparatus and System for Neural Network Hearing Aid,” filed Jan. 14, 2022, which is herein incorporated by reference in its entirety. (4) This application is a continuation-in-part, claiming the benefit under 35 U.S.C. § 120, of U.S. application Ser. No. 17/576,899, entitled “Method, Apparatus and System for Neural Network Hearing Aid,” filed Jan. 14, 2022, which is

herein incorporated by reference in its entirety. (5) This application is a continuation-in-part of International Patent Application Serial No. PCT/US2022/012567, entitled “Method, Apparatus and System for Neural Network Hearing Aid,” filed Jan. 14, 2022, which is herein incorporated by reference in its entirety. (6) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/305,676 filed Feb. 1, 2022, and entitled “SYSTEM AND METHOD FOR ENHANCING SPEECH OF TARGET SPEAKER FROM AUDIO SIGNAL IN AN EAR-WORN DEVICE USING VOICE SIGNATURES,” which is herein incorporated by reference in its entirety.

BACKGROUND

Field

(1) The present application relates to ear-worn speech enhancement devices.

Related Art

(2) Hearing aids are used to help those who have trouble hearing to hear better. Typically, hearing aids amplify received sound. Some hearing aids attempt to remove environmental noise from the incoming sound.

BRIEF SUMMARY

(3) Some embodiments provide for a method for selectively processing with an ear-worn device a target speaker's speech from an audio signal comprising the target speaker's speech and speech from additional speakers. The ear-worn device includes a processor and a microphone coupled to the processor. The method comprises: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and increasing, with the processor of the ear-worn device, a signal-to-noise ratio (SNR) of the target speaker's speech by processing the audio signal with a machine learning model using a voice signature of the target speaker.

(4) Some embodiments provide for an apparatus comprising a processor and a microphone coupled to the processor. The apparatus is an ear-worn device. The processor is configured to selectively process a target speaker's speech from an audio signal comprising the target speaker's speech and speech from additional speakers. The processing includes: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and increasing, with the processor of the ear-worn device, a signal-to-noise ratio (SNR) of the target speaker's speech by processing the audio signal with a machine learning model using a voice signature of the target speaker.

(5) Some embodiments provide for a method for selectively processing with an ear-worn device a target speaker's speech from an audio signal comprising the target speaker's speech and speech from additional speakers. The ear-worn device includes a processor and a microphone coupled to the processor. The method comprises: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and increasing, with the processor of the ear-worn device, a signal-to-noise ratio (SNR) of the target speaker's speech by processing the audio signal with a machine learning model using a voice signature of the target speaker.

(6) Some embodiments provide for an apparatus comprising a processor and a microphone coupled to the processor. The apparatus is an ear-worn device. The processor is configured to selectively process a target speaker's speech from an audio signal comprising the target speaker's speech and speech from additional speakers. The processing includes: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and increasing, with the processor of the ear-worn device, a signal-to-noise ratio (SNR) of the target speaker's speech by processing the audio signal with a machine learning model using a voice signature of the target speaker.

(7) Some embodiments provide for a method for operating a mobile processing device operatively

couplable to an ear-worn device. The method comprises: wirelessly transmitting, from the mobile processing device to the ear-worn device, a voice signature of at least one target speaker.

(8) Some embodiments provide for an apparatus comprising a processor, where the apparatus is operatively couplable to an ear-worn device. The apparatus is a mobile processing device. The processor is configured to: wirelessly transmit, from the mobile processing device to the ear-worn device, a voice signature of at least one target speaker.

(9) Some embodiments provide for a system comprising: a hearable device comprising a microphone configured to receive an audio signal comprising temporally overlapping speech components from multiple speakers; and at least one processor configured to process the audio signal received by the microphone to identify a target speaker among the multiple speakers.

(10) Some embodiments provide a method of selectively processing, with an ear-worn device including a processor and a microphone coupled to the processor, a target speaker's speech from an audio signal. The method comprises detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and isolating, with the processor of the ear-worn device, a component of the audio signal representing speech; determining, with the processor of the ear-worn device, that the component of the audio signal representing speech represents speech of a target or non-target speaker; and applying a relative gain to the audio signal in dependence on determining that the component of the audio signal represents speech of the target or non-target speaker.

(11) Some embodiments provide for non-transitory computer-readable medium including instructions that when executed by a processor, perform one or more of the methods listed above.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Various aspects and embodiments of the application will be described with reference to the following figures. It should be appreciated that the figures are not necessarily drawn to scale. Items appearing in multiple figures are indicated by the same reference number in all the figures in which they appear.

(2) FIG. 1 illustrates an example multi-speaker environment and an audio system including an ear-worn device and a separate electronic device, according to a non-limiting embodiment of the present application.

(3) FIG. 2 illustrates communication between an ear-worn device and a separate electronic device, according to a non-limiting embodiment of the present application.

(4) FIG. 3 illustrates a system with an ear-worn device and a portable electronic device for selectively enhancing speech from a target speaker, according to a non-limiting embodiment of the present application.

(5) FIG. 4A illustrates example components of an ear-worn device that may be configured to enhance speech of a target speaker in a multi-speaker environment, according to a non-limiting embodiment of the present application.

(6) FIG. 4B illustrates example components of a variation of the ear-worn device in FIG. 4A that may be configured to enhance speech of a target speaker in a multi-speaker environment, according to a non-limiting embodiment of the present application.

(7) FIGS. 5A-5B illustrate example components of an ear-worn device having two microphones, according to a non-limiting embodiment of the present application.

(8) FIGS. 6A and 6B illustrate an example configuration of a voice isolation network, according to a non-limiting embodiment of the present application.

(9) FIG. 6C illustrate an example signal stream that may be concatenated from an audio signal and voice signature(s) of target speaker(s), according to a non-limiting embodiment of the present

application.

(10) FIG. 7A is a flowchart of an example method of operation of an ear-worn device configured to selectively isolate speech from a target speaker within a multi-speaker environment, according to a non-limiting embodiment of the present application.

(11) FIG. 7B is a variation of the example method in FIG. 7A of operation of an ear-worn device configured to selectively isolate speech from a target speaker within a multi-speaker environment, according to a non-limiting embodiment of the present application.

(12) FIG. 8 is a block diagram illustrating training and deployment of a voice isolation machine learning model for isolating speech from a target speaker, according to a non-limiting embodiment of the present application.

(13) FIG. 9 illustrates a block diagram of a system-on-chip (SOC) package that may be implemented in an ear-worn device, according to a non-limiting embodiment of the present application.

(14) FIG. 10 is a block diagram illustrating a portion of a circuitry configuration of an electronic device operable to extract voice signature(s), according to a non-limiting embodiment of the present application.

(15) FIG. 11A illustrates an example graphical user interface that may be implemented in an electronic device to select one or more target speakers, according to a non-limiting embodiment of the present application.

(16) FIG. 11B illustrates a block diagram of an example process for implementing the example graphical user interface of FIG. 11A, according to a non-limiting embodiment of the present application.

(17) FIG. 11C illustrates an example graphical user interface that may be implemented in an electronic device to select one or more target speakers, according to a non-limiting embodiment of the present application.

(18) FIG. 11D illustrates a block diagram of an example process for implementing the example graphical user interface of FIG. 11C, according to a non-limiting embodiment of the present application.

(19) FIG. 11E illustrates an example graphical user interface that may be implemented in an electronic device to add a new speaker to a registry of known speakers, according to a non-limiting embodiment of the present application.

(20) FIG. 11F illustrates a block diagram of an example process for implementing the example graphical user interface of FIG. 11E, according to a non-limiting embodiment of the present application.

(21) FIG. 12 illustrates a block diagram of an example process for collecting a voice signature of a speaker, according to a non-limiting embodiment of the present application.

(22) FIG. 13 is a block diagram illustrating training and deployment of a voice signature machine learning model for extracting voice signature(s) from speech data, according to a non-limiting embodiment of the present application.

(23) FIG. 14 illustrates an example of a computing system that may be implemented in an electronic device to implement various embodiments described in the present application.

(24) FIG. 15 illustrates an example circuit including a voice isolation network and a voice signature network, according to a non-limiting embodiment of the present application.

(25) FIG. 16 illustrates an example of a voice isolation network having a voice isolation model for de-noising an input audio signal and determining an embedding of the input audio signal.

(26) FIG. 17 illustrates a voice isolation and classification network according to some embodiments of the present technology.

DETAILED DESCRIPTION

(27) Aspects of the present technology provide hearing systems and methods for de-noising a received audio signal, identifying the presence of a target or non-target speaker's speech in the

received audio signal, and processing the received audio signal for output to a listener based on detection of the target or non-target speaker's speech. The hearing system may include an ear-worn device, such as a hearing aid, and a separate electronic device, such as a mobile phone or tablet, in communication with the ear-worn device. The target speaker may be one or more conversation partners of the wearer of the ear-worn device, while a non-target speaker may be another conversation partner or may be the wearer of the ear-worn device herself. The ear-worn device may process the received audio signal using a machine learning model. In some embodiments, the machine learning model also receives a voice signature of the target and/or non-target speaker(s). The voice signature may be used by the machine learning model to identify the speech of the target and/or non-target speaker(s), and the hearing system may then preferentially process the speech of the target and/or non-target speaker(s). In some embodiments, the machine learning model may process the received audio signal by de-noising the audio signal and determining an embedding of the audio signal, which may be compared to a reference embedding representing a voice signature. The audio signal may then be processed differently depending on whether the embedding determined from the received audio signal matches the reference embedding. For example, the wearer's own voice may be attenuated, or the speech of a conversation partner may be enhanced.

(28) According to an aspect of the technology described herein, an ear-worn device is provided that operates to isolate and individually treat the received speech of a target speaker or multiple target speakers from an audio input signal detected in a multi-speaker environment. The ear-worn device, which is a hearing aid in some embodiments, uses a machine learning model that receives a voice signature of each of one or more target speakers as input signals, to identify and isolate the component of the audio input signal attributable to the target speaker(s). Once isolated, the target speaker's speech may be enhanced, de-emphasized, or otherwise processed in a manner desired by the wearer of the ear-worn device. As a result, the wearer of the ear-worn device can have a positive experience in multi-speaker environments.

(29) The inventors have recognized that conventional hearing aids do not perform well in multi-speaker environments. Some hearing aids amplify all received sounds. Some hearing aids attempt to filter out ambient noise and amplify all speech received. The inventors have appreciated that such approaches perform inadequately in some hearing aid use scenarios including multi-speaker environments, which is exacerbated by poor performance at filtering out ambient noise. Examples of multi-speaker environments include family gatherings, meals with multiple people, conference meetings, networking events, playground settings, and school classrooms. In these and other multi-speaker environments, an individual often wishes to listen to a subset of the speakers present. For instance, multiple conversations may occur simultaneously between different people seated at a table, and a hearing aid wearer may wish to pay attention to one of the conversations and not the others. Hearing aids that simply amplify all sound or that amplify all speech are inadequate in such a setting, as they are in other multi-speaker environments, because they fail to provide the user with the desired level of hearing focus.

(30) The inventors have also recognized that conventional hearing aids do not perform well at reducing or excluding the wearer's own voice. As described above, some hearing aids amplify all received sounds, including the wearer's own speech. Such behavior can worsen the wearer's experience. People are used to hearing their own voice both traveling through the air into their ears and traveling through the bones of their head to their ears, with both signals arriving at very low latency. The bone conduction path typically has a different frequency characteristic than the air conduction path. When also played back through a hearing aid, an amplified version of the person's voice is typically at a delay of a few milliseconds and has the frequency characteristics of the air conduction path. Hearing aid wearers typically find the experience of hearing themselves more loudly, with an uncharacteristic sound, to be initially very unnatural. As a result, people can be dissuaded from wearing a hearing aid even when they have poor hearing.

(31) The consequences of poor hearing aid performance are significant. Hearing is a fundamental

sense which impacts how people experience their settings and interactions with others. Poor hearing aid experience leads to the hearing aid user withdrawing from those activities in which the hearing aid does not work well or avoiding such activities altogether. Withdrawing from these common and innately human multi-speaker environments can be detrimental to family connections and friendships, among other things, and can contribute to the further health and emotional decline of the individual hard of hearing.

(32) Aspects of the present application provide hearing aids or other ear-worn devices that provide a wearer with a positive experience in multi-speaker environments by aiding the wearer in focusing on the speech of a desired subset of the speakers in the multi-speaker environment. For example, the speech of one or more target speakers may be selectively amplified. The speech of one or more non-target speakers may be reduced or eliminated. For instance, the ear-worn device wearer may not want to hear his or her own voice, and therefore may select for the ear-worn device to deemphasize his or her own speech. In at least some embodiments, the wearer can select the target speaker(s) or non-target speaker(s).

(33) Aspects of the present application provide an intelligent ear-worn device that provides a true audio experience by selectively isolating speech from one or more target speakers in a multi-speaker environment without noticeable delay of the audio signal containing the speech. The ear-worn device may feature a temporal lobe on a chip executing a machine learning model that analyzes an incoming audio signal, identifies and isolates speech in the incoming audio signal attributable to the target speaker, and presents that speech to the wearer. The isolated speech is presented to the wearer within a time amounting to no more than an insignificant delay from the perspective of the wearer, as would not negatively impact natural conversation. Thus, the machine learning model operates to identify and isolate the speech within that time.

(34) According to an aspect of the technology described herein, the input audio signal may be segmented into small samples (segments), which are processed sequentially. An audio segment may be provided to the machine learning model to generate isolated speech. While the isolated speech is further processed and played back to the wearer of the ear-worn device, the subsequent audio segment is being provided to and processed by the machine learning model. Thus, when the playback of the current audio segment is completed, the subsequent audio segment outputted by the machine learning model will be ready for further processing and playback. Such technology, combined with a choice of machine learning model, for example, a recurrent neural network, may facilitate the ear-worn device processing the speech signal with a machine learning model without introducing noticeable delay to the wearer's ears.

(35) The inventors have recognized that voice signatures may be used by ear-worn devices to isolate the speech of one or more target speakers or non-target speakers. Individual speakers typically exhibit unique speech characteristics. The unique speech characteristics can be used to uniquely identify the respective speaker, and thus may serve as a voice signature. The voice signature may take various forms usable by a processor of an ear-worn device. The ear-worn device may process the voice signature in combination with an audio signal received by the ear-worn device to identify and isolate the component of the audio signal attributable to the speaker associated with the voice signature. The processing may be performed using a machine learning model executing on the ear-worn device.

(36) According to an aspect of the technology described herein, an ear-worn device is configured to use a machine learning model that operates on a voice signature and an input audio signal containing speech from multiple speakers to isolate a component of the speech representing speech of a target speaker associated with the voice signature. The machine learning model receives the input audio signal as one input signal and a voice signature of the target speaker as a second input signal, and outputs the isolated speech component. The voice signature is a feature vector including frequency domain components. The voice signature may be used by the machine learning model to generate a complex ideal ratio mask that may be applied to the input audio signal to isolate the

target speaker's speech from the input audio signal. After isolating the target speaker's speech, that speech may be preferentially treated (e.g., amplified) to produce an enhanced output audio signal for the ear-worn device.

(37) According to an aspect of the technology described herein, the voice signatures used by the machine learning model of an ear-worn device may be provided by a separate electronic device that, together with the ear-worn device, forms an audio system. The separate electronic device may be a smartphone, tablet computer, personal digital assistant (PDA), or other device in communication with the ear-worn device. The separate electronic device may provide one or more voice signatures to the ear-worn device, for example in response to user selection. For example, the separate electronic device may contain a registry of voice signatures from which the user of the ear-worn device may select. The provided voice signature(s) may serve as an input to the machine learning model executed by the ear-worn device to isolate speech attributable to the speaker associated with the voice signature(s).

(38) Thus, according to an aspect of the technology described herein, a registry of voice signatures is provided. The voice signature registry includes one or more voice signatures associated with known speakers, who are potential target speakers and non-target speakers for an ear-worn device wearer. One or more signatures may be selected from the registry for use in a machine learning model of an ear-worn device, to allow the ear-worn device to identify and isolate speech attributable to speakers associated with the selected voice signature(s). In some embodiments, the machine learning model may receive a subset of the voice signatures associated with the registry representing selected known speakers that are present in a given conversation.

(39) The voice signature registry may be stored in various locations. In one embodiment, the voice signature(s) are stored on the external electronic device (e.g., the smartphone). In some embodiments, a user of the external electronic device may be the wearer of the ear-worn device. In a given conversation (e.g., in a multi-speaker environment), a user of the external electronic device may select the target speaker(s) on the external electronic device from among known speakers in the registry, e.g., from an option list (menu). Alternatively, and/or additionally, the external electronic device may automatically identify target speakers by determining whether the input audio signal in a conversation includes speech components of one or more known speakers in the registry. The user may then be presented with a menu of the identified target speakers from which to select. Once the target speaker(s) are identified and/or selected, the voice signature(s) associated with the target speaker(s) are sent to the ear-worn device. In one embodiment, the voice signature(s) are stored on the ear-worn device itself and can be called from memory. In such case, instead of transmitting the voice signature(s) of the target speaker(s) to the ear-worn device, the external electronic device may transmit identifier(s) of the target speaker(s) to the ear-worn device, which in turn can retrieve the stored voice signature(s) of target speaker(s) based on the identifiers.

(40) The voice signatures associated with the registry of known speakers may be collected in various ways. In some embodiments, the voice signatures may be extracted from audio input using another machine learning model, e.g., a voice signature machine learning model implemented in a voice signature network separate from the machine learning model used by the ear-worn device to isolate speech from a received audio signal. A sample of speech from the target speaker may be provided as input to the voice signature network. The voice signature network may output a multi-dimensional feature vector representing the voice signature for that target speaker. The machine learning model used to extract voice signatures may operate on the separate electronic device (e.g., the smartphone) in some embodiments. In some other embodiments, the machine learning model used to extract voice signatures may operate on the ear-worn device.

(41) The sample speech of the target speaker used by the voice signature network to extract the voice signature may be obtained in various manners. In some examples, input audio signal including the sample speech may be detected using the microphone of the external electronic device. In other examples, the input audio signal including the sample speech may be detected

using the microphone of the ear-worn device, and the ear-worn device then transmits the input audio signal to the external electronic device. The input audio signal may be transmitted to the external electronic device wirelessly. In some embodiments, the target speaker may be provided a microphone and may read a predetermined script. Alternatively, the target speaker may provide a speech sample online. In a further alternative, the speech may be dynamically extracted from audio input to the ear-worn device wearer. For instance, while having a conversation with a given speaker, the ear-worn device may collect an audio sample and provide that sample to the voice signature network, which may extract the voice signature.

(42) In some embodiments, multiple target speakers may be identified in a multi-speaker conversation. In the multi-speaker conversation, multiple target speakers may speak at different or overlapping times. In some embodiments, the ear-worn device is configurable to isolate the speech of multiple target speakers. The neural network may receive multiple voice signatures as input(s) and analyze the incoming audio signal using those voice signatures to isolate speech components from the input audio signal attributable to the multiple target speakers.

(43) As should be appreciated from the foregoing, according to some embodiments of the technology described herein, an ear-worn device is configured to use a voice isolation machine learning model that operates on a voice signature to isolate a component of the speech representing speech of a target speaker associated with the voice signature. The voice signature may be obtained from a separate machine learning model, such as a voice signature machine learning model, configured to extract the voice signature from one or more reference audio clips representing speech of the target or non-target speaker. Such an embodiment may be advantageous for multiple reasons. As a non-limiting example, the voice signature machine learning model may be implemented on an electronic device different from the ear-worn device, thereby reducing the computational complexity of implementing multiple machine learning models on the ear-worn device, while retaining voice isolation capabilities.

(44) According to an aspect of the technology, an ear-worn device is configured to use a voice isolation and classification machine learning model that operates on an input audio signal to both (a) de-noise the input audio signal and (b) determine an embedding of the input audio signal. The voice isolation and classification machine learning model may be used to determine an embedding representative of a target or non-target speaker, for example by supplying the voice isolation and classification machine learning model with a sample audio clip from the target or non-target speaker. Subsequently, the voice isolation and classification model may operate on input audio signals to de-noise the input audio signal and determine an embedding of the input audio signal, and the determined embedding may be compared to the embedding determined previously to correspond to the target or non-target speaker. The manner of processing the de-noised audio signal may depend on whether the two embeddings are found to match. For example, if the non-target speaker is the hearing aid wearer, then the de-noised audio signal may be attenuated when the two embeddings are determined to match, whereas the de-noised audio signal may be amplified if the two embeddings do not indicate a match.

(45) As described above, according to an aspect of the present application, the ear-worn device may be configured to suppress the wearer's own voice. Using a voice signature of the wearer, the machine learning model may suppress such audio. The wearer of the ear-worn device may select such operation when he or she wishes to only hear the speech of his or her conversation partners.

(46) The aspects and embodiments described above, as well as additional aspects and embodiments, are described further below. These aspects and/or embodiments may be used individually, all together, or in any combination of two or more, as the disclosure is not limited in this respect.

(47) As described above, aspects of the technology described herein provide an ear-worn device that operates to isolate and individually treat the received speech of a target speaker or multiple target speakers from an audio input signal detected in a multi-speaker environment. FIG. 1

illustrates a multi-speaker environment and an audio system including an ear-worn device of the type described and a separate electronic device. The multi-speaker environment **100** includes ear-worn device wearer **102** and other individuals, including a target speaker **104** and two non-target speakers **106a** and **106b**. The ear-worn device wearer **102** has an ear-worn device **108** and a separate electronic device **110**. The ear-worn device **108** and electronic device **110** in combination represent an audio system.

(48) The illustrated setting of the multi-speaker environment **100** is one in which multiple individuals may speak simultaneously. In the illustrated example, the individuals are seated around a table. For instance, the illustrated individuals may be having a meeting, playing a game, or may be having a meal. Other scenarios are possible as well, as the specific setting is not limiting of the various aspects of the technology described herein. In this context, multiple of the illustrated individuals may be speaking at the same time. For example, the ear-worn device wearer **102** and the target speaker **104** may be having a conversation, and the non-target speakers **106a** and **106b** may be having a conversation. There may be other reasons that multiple of the illustrated individuals may be speaking simultaneously.

(49) The ear-worn device wearer **102** is wearing an ear-worn device **108** which detects sound and outputs an audio signal to the ear-worn device wearer **102**. For example, the ear-worn device wearer **102** may be hard of hearing, and the ear-worn device **108** may be a hearing aid. The ear-worn device **108** may be capable of detecting environment sound as well as the speech of the target speaker **104** and non-target speakers **106a** and **106b**. The ear-worn device wearer, however, may wish to listen to the target speaker **104**, and not the non-target speakers **106a** and **106b**. The ear-worn device **108** may detect the speech of all the illustrated speakers, but output to the ear-worn device wearer **102** an audible output signal representing the speech of the target speaker **104**, with the speech of non-target speakers **106a** and **106b** reduced or eliminated.

(50) The ear-worn device **108** and the electronic device **110** may work in combination to allow the ear-worn device wearer **102** to listen to only the speech of target speaker **104**. The electronic device **110** may store a registry of voice signatures. For example, the electronic device **110** may store voice signatures for each of the target speaker **104**, non-target speaker **106a**, and non-target speaker **106b**. When the ear-worn device wearer **102** wants to listen to only target speaker **104**, the ear-worn device wearer **102** may select the voice signature for target speaker **104** from the voice signature registry on electronic device **110**. The electronic device **110** may provide the voice signature to the ear-worn device **108**, which may use the provided voice signature to identify and isolate speech attributable to target speaker **104** from a detected audio signal containing speech components attributable to target speaker **104** and at least one of non-target speaker **106a** or non-target speaker **106b**.

(51) FIG. 2 illustrates an example implementation of the audio system illustrated in FIG. 1. As shown, the audio system **200** includes the ear-worn device **108** and the electronic device **110**. In this example, the ear-worn device is a hearing aid and the electronic device **110** is a smartphone. The electronic device **110** includes a display screen **202** which can display entries **204** of a voice registry. The ear-worn device wearer can select the entry of interest to be the target speaker or a speaker the ear-worn device wearer would prefer to selectively not hear. The voice signature(s) corresponding to the selected speaker(s) can be sent to the ear-worn device **102** via a wireless communication link **206**. The ear-worn device **108** may use the received voice signature from the electronic device to process audio signals detected from the microphone(s) of the ear-worn device to isolate speech attributable to the selected speaker(s). The isolated speech may be output to the ear-worn device wearer **102** through the speaker device(s) (e.g., receiver(s)) of the ear-worn device.

(52) FIG. 3 illustrates a system with an ear-worn device and a portable electronic device for selectively enhancing speech from a target speaker, according to a non-limiting embodiment of the present application. Audio system **300** may be an example implementation of the system shown in FIGS. 1 and 2. For example, audio system **300** may include an ear-worn device **302** and electronic

device **304**. The ear-worn device **302** may be an example implementation of the ear-worn device **108** of FIGS. **1** and **2**. Ear-worn device **302** as described in FIG. **3** may have various forms. For example, the ear-worn device may be a hearing aid or a headphone, or any suitable wearable audio device. Additionally, ear-worn device **302** may include a communication port **314** configured to communicate (e.g., wired or wirelessly) with an external device and exchange data with the external device, such as electronic device **304**. Electronic device **304** may be an example implementation of the electronic device **110** of FIGS. **1** and **2**. For example, electronic device **304** may be a smart phone, or any suitable portable electronic device associated with the wearer of the ear-worn device.

(53) In some non-limiting examples, ear-worn device **302** may include a microphone **308** and a speaker device (e.g., a receiver) **312**. Microphone **308** may be configured to detect audio signal **336** from sound (e.g., speech). For example, the audio signal may include temporally overlapping speech components from multiple speakers. Ear-worn device **302** may be capable of processing the audio signal **336** detected by the microphone **308** to isolate a component of the audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers. In some embodiments, ear-worn device **302** may be configured to process the audio signal **336** with a voice isolation network using a voice signature of the target speaker. In some examples, the voice signature of a speaker may be a multi-dimensional feature vector. The voice signature may contain data in frequency-domain, time-domain, or any suitable data that may be representative of different speakers. Receiver **312** may include an audio output device configured to playback the output from the voice isolation network to the ear-worn device wearer, such as a speaker device.

(54) Voice signature of a speaker may be a unique digital representation of a speaker, where voice signatures of different speakers are distinctive. In some embodiments, the system may use a trained voice signature network to extract the voice signature of a speaker from an audio signal containing speech of the speaker. When the voice signature network is properly trained, the voice signatures for different speakers (i.e., speaker embeddings) extracted using the voice signature network may be distinctive and the voice signatures extracted from different audio signal including speech of the same speaker may be the same. Using the voice signature that is unique to a selected target speaker, the voice isolation network may isolate the speech components from an audio signal, where the isolated speech component is attributable to the selected target speaker. In some embodiments, each of the voice signature network and the voice isolation network may include a different machine learning model. For example, the voice signature network may include a voice signature machine learning model configured to receive an audio signal as input and output voice signature of the speaker whose speech is included in the audio signal. The voice isolation network may include a voice isolation machine learning model configured to receive two inputs, an audio signal including a target speaker(s) and a voice signature(s) of the target speaker(s), and output isolated speech of the target speaker(s). Details of the voice isolation network will be further described in with reference to FIGS. **6-8**. Details of the voice signature network will be further described with reference to FIGS. **12** and **13**.

(55) With continued reference to FIG. **3**, in some embodiments, ear-worn device **302** may store the voice signatures of target speakers locally, such as in a memory **316** containing the voice signatures of one or more speakers. In some embodiments, ear-worn device **302** may receive the voice signature(s) of target speaker(s) from an external device, such as electronic device **304**. For example, ear-worn device **302** may be configured to communicate wirelessly with electronic device **304** in a multi-speaker environment, e.g., a multi-speaker conversation as shown in FIG. **1**. Ear-worn device **302** may be configured to transmit a segment of an audio signal **332** detected from the microphone of the ear-worn device to electronic device **304**. Alternatively, a microphone of the electronic device **304** may be configured to detect the audio signal **336** from the conversation. The segment of the audio signal may be of a length, e.g., a few seconds to a few minutes, and includes

speeches from multiple speakers in the multi-speaker conversation, e.g., at the beginning of the conversation.

(56) Electronic device **304** may be configured to process the audio signal **336** detected by the microphone of the ear-worn device (or the electronic device itself) to identify one or more speakers in the conversation. In some embodiments, in identifying the one or more speakers in the conversation, electronic device **304** may be configured to execute a voice signature network described above. The voice signature network may be configured to extract one or more voice signatures as output from input audio signal containing speech component(s) of one or more target speakers. Electronic device **304** may be configured to further identify one or more speakers in the conversation based on the extracted voice signature(s), against known speakers. In some embodiments, electronic device **304** may include a registry **320** of known speakers stored in a memory of the electronic device. Electronic device **304** may also include voice signatures associated with known speakers pre-stored in the memory.

(57) In some embodiments, electronic device **304** may identify one or more speakers in the conversation by matching the extracted voice signature(s) to the voice signatures of known speakers in the registry **320**. Electronic device **304** may automatically select target speaker(s) from the registry of known speakers whose signatures are matched to the extracted voice signature, assuming the target speakers are known speakers (of the wearer of the ear-worn device) that are speaking in the conversation. Additionally, and/or alternatively, electronic device **304** may receive user selection(s) identifying the target speakers from the identified speakers. For example, the user may select a subset of the previously identified speakers being matched and whom the wearer of the ear-worn device prefers to listen. In some embodiments, without executing the voice signature network, the electronic device **304** may display a list of known speakers in the registry and receives a user selection indicating a selection of one or more known speakers in the register whom the wearer of the ear-worn device knows is/are present in the conversation and prefers to listen to. In some other embodiments, if an extracted voice signature does not match to any voice signatures associated with the registry of known speakers, electronic device **304** may add a new speaker to the registry. The user may select that speaker as the target speaker, and along with additional other target speakers.

(58) Subsequently, electronic device **304** may transmit the voice signature(s) of the selected target speaker(s) **334** to the ear-worn device **302**. In some examples, electronic device **304** may transmit the voice signature(s) of the selected target speaker(s) **334** to the ear-worn device **302**.

Alternatively, as described above, the ear-worn device **302** may store the voice signatures of multiple speakers. In such case, electronic device **304** may transmit identifier(s) of the selected target speaker(s) to ear-worn device **302**, which in turn can retrieve the stored voice signature(s) of target speaker(s) based on the identifiers.

(59) In some examples, electronic device **304** may also communicate (wired or wirelessly) with one or more servers **306**, via a communication network, to cause the server(s) **306** to perform some of the operations described above. In a non-limiting example, electronic device **304** may communicate with server **306** to cause server **306** to perform extraction of the voice signature. In such case, the electronic device **304** may provide input audio signal of a conversation to the server **306**. Server **306** may include a voice signature network, which may be triggered by the electronic device to extract voice signature(s) from the input audio signal.

(60) FIG. 4A illustrates example components of an ear-worn device that may be configured to enhance speech of a target speaker in a multi-speaker environment, according to a non-limiting embodiment of the present application. In some embodiments, ear-worn device **400** may be an implementation of at least a portion of the ear-worn device **108** of FIGS. 1 and 2 and **300** of FIG. 3. Ear-worn device **400** may include one or more microphones **402**, one or more receivers **405**, and a voice isolation network **403** coupled in between the microphone(s) **402** and the receiver(s) **405**. In some embodiments, microphone(s) **402** may be configured to detect audio signal. The audio signal

may be generated by the microphone(s) from sound **401**, e.g., speech in a conversation. In a multi-speaker conversation, the audio signal detected by the microphone(s) may include speech components attributable to multiple speakers. In some embodiments, the audio signal detected by the microphone(s) may be analog signal. The ear-worn device **400** may additionally include an analog-to-digital converter (ADC, not shown) to convert the analog signal to digital signal **406** as input to the voice isolation network **403**. In some embodiments, the microphone(s) **402** may be capable of producing digital audio signals. In such case, the audio signal detected by the microphone(s) may be digital signal **406**, which can be directly provided to the voice isolation network **403**.

(61) With further reference to FIG. **4A**, voice isolation network **403** may receive the digital audio signal **406** and process the digital audio signal **406** to output isolated speech **407**. Receiver(s) **405** may be configured to output the isolated speech **407** for playback to the wearer of the ear-worn device. For example, the receiver(s) **405** may receive the digital signal **407** from the voice isolation network and convert the digital signal **407** to analog signal before producing the output signal **409**. The receiver **405** may be a speaker device (e.g., loudspeaker) in some embodiments. In other examples, the ear-worn device may additionally include a digital-to-analog converter (DAC, not shown) to convert the digital signal **407** to analog signal as input to the receiver(s) **405** for providing the output signal **409**.

(62) In some embodiments, ear-worn device **400** may include a digital signal processor (DSP, **404**) coupled between the voice isolation network **403** and the receiver(s) **405**. The DSP **404** may be configured to process the isolated speech from the voice isolation network **403** and generate an enhanced output **408**. For example, DSP **404** may include a frequency-based amplification. In some embodiments, the isolated speech output from the voice isolation network may include preferentially processed (e.g., amplified or suppressed) speech components attributable to target/non-target speakers. For example, the output from the voice isolation network may include speech with increased signal-to-noise ratio (SNR) for the target speaker's speech, or the volume of one or more target speaker(s) at a desirable level selected by the user (e.g., the wearer of the ear-worn device). The details of preferentially treating the speech attributable to target/non-target speakers will be further described in embodiments of a voice isolation network with reference to FIGS. **6A-6B**.

(63) FIG. **4B** illustrates example components of a variation of the ear-worn device in FIG. **4A** that may be configured to enhance speech of a target speaker in a multi-speaker environment, according to a non-limiting embodiment of the present application. In some embodiments, ear-worn device **470** may be an implementation of at least a portion of the ear-worn device **108** of FIGS. **1** and **2** and **300** of FIG. **3**. Ear-worn device **470** may have microphone(s) **420** to receive one or more audio input signal **410** and receiver(s) **460**, similar to microphone(s) **402** and receiver(s) **405**, respectively, described in FIG. **4A**. The receivers **460** may process the output signal **445** and output an output signal **490**. The receivers **405** and **460** may be considered an output module or output block in some embodiments. Ear-worn device **470** may also include voice isolation network **450** similar to voice isolation network **403** of FIG. **4A**, and digital signal processor (DSP, **440**) similar to DSP **404** of FIG. **4A**. Additionally, ear-worn device **470** may include controller **430** configured to control both the voice isolation network **450** and DSP **440**.

(64) Controller **430** receives digital audio signal **425**. Controller **430** may comprise one or more processor circuitries (herein, processors), memory circuitries and other electronic and software components configured to, among others, (a) perform digital signal processing manipulations necessary to prepare the signal for processing by the voice isolation network **450** or the DSP **440**, and (b) to determine the next step in the processing chain from among several options. In one embodiment of the disclosure, controller **430** executes a decision logic to determine whether to advance signal processing through one or both of DSP **440** and voice isolation network **450**. For example, DSP **440** may be activated at all times, whereas controller **430** executes decision logic to

determine whether to activate the voice isolation network **450** or bypass the voice isolation network by deactivating the voice isolation network **450**. In some embodiments, DSP **455** may be configured to apply a set of filters to the incoming audio components. Each filter may isolate incoming signals in a desired frequency range and apply a non-linear, time-varying gain to each filtered signal. The gain value may be set to achieve dynamic range compression or may identify stationary background noise. DSP **440** may then recombine the filtered and gained signals to provide an output signal **445**.

(65) The controller **430** may include storage circuitry **432** to store data, such as data representing voices that, when detected, may serve as an input to the controller's logic. For example, the storage circuitry **432** may include a speaker registry of the types described herein, in those embodiments in which the voice registry is stored on the ear-worn device.

(66) As stated, in one embodiment, the controller performs digital signal processing operations to prepare the signal for processing by one or both of DSP **440** and voice isolation network **450**. Voice isolation network **450** and DSP **440** may accept as input the signal in the time-frequency domain (e.g., signal **425**), so that controller **430** may take a Short-Time Fourier Transform (STFT) of the incoming signal before passing it onto either voice isolation network **450** or DSP **440**. In another example, controller **430** may perform beamforming of signals received at different microphones to enhance the audio signals coming from certain directions.

(67) In certain embodiments, controller **430** continually determines the next step in the signal chain for processing the received audio data. For example, controller **430** activates voice isolation network **450** based on one or more of user-controlled criteria, user-agnostic criteria, user clinical criteria, accelerometer data, location information, stored data and the computed metrics characterizing the acoustic environment, such as SNR. For example, in response to a determination that the speech is continual, or that the SNR of the input audio signal is above a threshold ratio, controller **430** may activate the voice isolation network. Otherwise, controller **430** may deactivate the voice isolation network **450**, leaving the DSP **440** activated. This results in a power saving of the ear-worn device when the voice isolation network is not needed. If voice isolation network **450** is not activated, controller **430** instead passes signal **435** directly to DSP **440**. In some embodiments, controller **430** may pass data to both voice isolation network **450** and DSP **440** simultaneously as indicated by arrows from controller **430** to DSP **440** and to voice isolation network **450**.

(68) In some embodiments, user-controlled criteria may represent one or more logics (e.g., hardware- or software-implemented). In some examples, user-controlled criteria may comprise user inputs including the selection of an operating mode through an application on a user's smartphone or input on the ear-worn device (for example by the wearer of the ear-worn device tapping the device). For example, when a user is at a restaurant, she may change the operating mode to noise cancellation/speech isolation by making an appropriate selection on her smartphone. Additionally, and/or alternatively, user-controlled criteria may comprise a set of user-defined settings and preferences which may be either input by the user through an application (app) or learned by the device over time. For example, user-controlled criteria may comprise a user's preferences around what sounds the wearer of the ear-worn device hears (e.g., new parents may want to always amplify a baby's cry, or a dog owner may want to always amplify barking) or the user's general tolerance for background noise. Additionally, and/or alternatively, user clinical criteria may comprise a clinically relevant hearing profile, including, for example, the user's general degree of hearing loss and the user's ability to comprehend speech in the presence of noise.

(69) User-controlled logic may also be used in connection with or aside from user-agnostic criteria (or logic). User-agnostic logic may consider variables that are independent of the user. For example, the user-agnostic logic may consider the hearing aid's available power level, the time of day or the expected duration of the voice isolation network operation (as a function of the anticipated voice isolation network execution demands).

(70) In some embodiments, acceleration data as captured on sensors in the device may be used by controller **430** in determining whether to direct signal controller output signal **435** to one or both of DSP **440** and voice isolation network **450**. Movement or acceleration information may be used by controller **430** to determine whether the user is in motion or sedentary. Acceleration data may be used in conjunction with other information or may be overwritten by other data. Similarly, data from sensors capturing acceleration may be provided to the voice isolation network as information for inference.

(71) In other embodiments, the user's location may be used by controller **430** to determine whether to engage one or both of DSP **440** and voice isolation network **450**. Certain locations may require activation of voice isolation network **450**. For example, if the user's location indicates high ambient noise (e.g., the user is strolling through a park or is attending a concert) and no direct conversation, controller **430** may activate DSP **440** only and deactivate voice isolation network **450**. On the other hand, if the user's location suggests that the user is traveling (e.g., via car or train) and other indicators suggest human communication, then controller **430** may activate voice isolation network **450** to enhance the audio signal by amplifying human voices over the surrounding noise.

(72) In some embodiments, controller **430** may execute an algorithmic logic to select a processing path. For example, controller **430** may detect SNR of input audio signal **425** and determine whether one or both of DSP **440** and voice isolation network **450** should be engaged. In one implementation, controller **430** compares the detected SNR value with a threshold value and determines which processing path to initiate. The threshold value may be one or more of empirically determined, user-agnostic or user-controlled. Controller **430** may also consider other user preferences and parameters in determining the threshold value as discussed above.

(73) In another embodiment, controller **430** may compute certain metrics to characterize the incoming audio as input for determining a subsequent processing path. These metrics may be computed based on the received audio signal. For example, controller **430** may detect periods of silence, knowing that silence does not require the voice isolation network to enhance and it should therefore deactivate the voice isolation network. In another example, controller **430** may include a Voice Activity Detector (VAD) **434** to determine the processing path in a speech-isolation mode. In some embodiments, the VAD may be a compact (e.g., much less computationally intensive) neural network in the controller.

(74) In an exemplary embodiment, controller **430** may receive the output of voice isolation network **450** for recently processed audio, as indicated by arrow from voice isolation network **450** to controller **430**, as input to controller **430**. Voice isolation network **450**, which may be configured to isolate target audio in the presence of background noise, provides the inputs necessary to robustly estimate the SNR. Controller **430** may in turn use the output of the voice isolation network **450** to detect when the SNR of the incoming signal is high enough or too low to influence the processing path. In still another example, the output of voice isolation network **450** may be used to improve the robustness of VAD **434**. Voice detection in the presence of noise is computationally intensive. By leveraging the output of voice isolation network **450**, ear-worn device **470** can implement this task with minimal computation overhead when the noise is suppressed based on isolated speech from the voice isolation network.

(75) When controller **430** utilizes voice isolation network output **451**, it can only utilize the output to influence the signal path for subsequently received audio signal. When a given sample of audio signal is received at the controller, the output of voice isolation network **450** for that sample will be computed with a delay, where the output of the voice isolation network, if computed before the next sample arrives, will influence the controller decision for the next sample. When the time interval of the sample is small enough, e.g., a few milliseconds or less than a second, such delay will not be noticeable by the wearer.

(76) When voice isolation network **450** is activated, using the output **451** of the voice isolation network **450** in the controller does not incur any additional computational cost. In certain

embodiments, controller **430** may engage voice isolation network **450** for supportive computation even in a mode when voice isolation network **450** is not the selected signal path. In such a mode, incoming audio signal is passed directly from controller **430** to DSP **440** but data (i.e., audio clips) is additionally passed at less frequent intervals to voice isolation network **450** for computation. This computation may provide an estimate of the SNR of the surrounding environment or detect speech in the presence of noise in substantially real time. In an exemplary implementation, controller **430** may send a 16 ms window of data once every second for VAD **134** detection at voice isolation network **150**. In some embodiments, voice isolation network **450** may be used for VAD **434** instead of controller **430**. In another implementation, controller **430** may dynamically adjust the duration of the audio clip or the frequency of communicating the audio clip as a function of the estimated probability of useful computation. For example, if the audio signal (e.g., **425**) exhibits a highly variable SNR, controller **430** may request additional voice isolation network computation at more frequent intervals.

(77) With reference to FIGS. **4A** and **4B**, ear-worn devices **400** and **470** may each include a single ear-piece having a microphone. In other examples, ear-worn devices **400** and **470** may each be binaural and include two ear-pieces, each ear-piece having a respective microphone. Similarly, ear-worn devices **400** and **470** may each include one or more receivers respectively included in one or two ear-pieces.

(78) FIGS. **5A-5B** illustrate variations of example components of an ear-worn device having two microphones, according to a non-limiting embodiment of the present application. FIGS. **5A** and **5B** each includes a portion of a circuitry **500**, **545** in an example ear-worn device, respectively. In some embodiments, the portions of circuitry **500**, **545** may be implemented in ear-worn device **108** (in FIGS. **1** and **2**), **302** (in FIG. **3**), **400** (in FIG. **4A**) and **470** (in FIG. **4B**), where the ear-worn device is binaural. In FIG. **5A**, circuitry **500** may include a beamformer **530** configured to process audio signal **519**, **529** respectively detected from microphones **514** and **524** (e.g., left and right microphones respectively residing in one of two ear-pieces of the ear-worn device and configured to receive input audio signals **510** and **520**, respectively). In some embodiments, beamformer **530** may be implemented in controller **430** of FIG. **4B**. Beamformer **530** may generate an enhanced audio signal **532** that accounts for sounds from different directions as detected by microphones **514** and **524**. As described above, the audio signals **519**, **529** respectively detected by the microphones **514** and **524** may be digital signals. The output from the beamformer **530** may be digital signal as well. As shown in FIG. **5A**, the enhanced audio signal **532** may be provided to the voice isolation network **540** in the ear-worn device. The voice isolation network **540** may be similar to the voice isolation network described above, e.g., **403** in FIG. **4A** and **450** in FIG. **4B**. The output of the voice isolation network may be provided to the receivers of two ear-pieces.

(79) In some embodiments, each ear-piece may be configured to communicate with the other ear-piece and exchange audio signal with the other ear-piece. For example, beamformer **530** may be residing in a first ear-piece of an ear-worn device. The audio signal detected by the microphone of the other ear-piece may be transferred from the other ear-piece to the ear-piece in which the beamformer **530** is residing. The output of the voice isolation network **540**, or the output of the DSP (e.g., **404** in FIG. **4A**, **440** in FIG. **4B**) may be transferred back to the other ear-piece. It is appreciated that the two ear-pieces may be configured to communicate using any suitable protocol, such as near-field magnetic induction (NFMI) protocol, which allows for fast data exchange over short distances. Further, beamformer **530** may be optional, where a binaural audio stream may be detected from microphones **514** and **524**, and provided to the virtual isolation network **540** without using a beamformer.

(80) With reference to FIG. **5B**, the circuitry **545** may include several components similar to those described in connection with FIG. **5A**. Microphone **554** is the same as microphone **514** in some embodiments. Microphone **564** is the same as microphone **524** in some embodiments. Microphone **554** receives input audio signal **550** and microphone **564** receives input audio signal **560**.

Microphone **554** outputs audio signal **559**, while microphone **564** outputs audio signal **569**.

(81) The circuitry **545** may include two separate voice isolation networks **570** and **580** each residing in a respective ear-piece of the ear-worn device. In such case, each ear-piece of the ear-worn device may include an independent configuration such as configurations described in **400** of FIG. **4A**) or **470** of FIG. **4B**. The circuitry **545** may further include a reconciler **590** configured to receive the output of the voice isolation network or the output of the DSP (e.g., **404** of FIG. **4B** or **440** of FIG. **4B**) from the two ear-pieces. The reconciler **590** may be configured to reconcile the outputs from the two ear-pieces and provide reconciled signals to the two ear-pieces for playback. As described above, the two ear-pieces may be configured to communicate with each other. In some embodiments, the output of the voice isolation network or the DSP of each one ear-piece may be transferred to the other ear-piece for reconciliation by the reconciler **590**, and the reconciled audio signal may be transferred back from the other ear-piece. In some other embodiments, each of the two ear-pieces may combine the output of the voice isolation network from the other ear-piece with the output of its own voice isolation network into a combined output and provide the combined output for further audio processing. For example, the output of the voice isolation network in one ear-piece (e.g., a complex ratio mask) may be transmitted to the other ear-piece. The other ear-piece may combine the received output with the output of its own voice isolation network (e.g., taking the average of two complex ratio masks) for further audio processing (e.g., DSP).

(82) FIGS. **6A** and **6B** illustrate an example configuration of a voice isolation network, according to a non-limiting embodiment of the present application. Voice isolation network **600** may be implemented as voice isolation network **403** (in FIG. **4A**), **450** (in FIG. **4B**), **540** (in FIG. **5A**) and **570**, **580** (in FIG. **5B**), in some examples, and may be implemented in ear-worn device **108** (in FIGS. **1** and **2**), **302** (in FIG. **3**), **400** (in FIG. **4A**), and **470** (in FIG. **4B**). Voice isolation network **600** may include a voice isolation machine learning (ML) model **602**, which may be configured to receive two inputs: audio signal **620** and voice signature(s) of target speaker(s) **622**. The audio signal provided as input to the voice isolation machine learning model may be audio signal detected from microphone(s) of an ear-worn device. The voice signature(s) of target speaker(s) may be obtained from an external electronic device associated with the wearer of the ear-worn device. The operations of the external electronic device that may be performed to provide voice signatures of selected target speaker(s) for the wearer of the ear-worn device will be further described in detail with reference to FIGS. **10-13**.

(83) The voice isolation network **600** may additionally include one or more components (e.g., relative gain filter **612**, recombiner **614**) to process the isolated speech from the voice isolation machine learning model **602** and preferentially treat (e.g., amplify or suppress) speech attributable to target/non-target speakers to produce an enhanced output audio signal **626**. The output audio signal **626** may be provided to a DSP (e.g., see FIGS. **4A** and **4B**) for outputting to the receiver(s) of the ear-worn device.

(84) Returning to FIG. **6A**, the input audio signal provided to the voice isolation machine learning model **602** may include speeches of multiple speakers (including target and non-target speakers) in a conversation. The audio signal may also include other signals, such as background noise. The voice signature(s) may include data (e.g., feature vectors) representative of selected target speakers in the multi-speaker conversation. In some examples, each voice signature feature vector (e.g., in frequency domain) may be a 256-dimensional embedding. Other suitable dimensions may also be possible. With reference to FIG. **6B**, voice isolation machine learning model **602** may include a combiner **604** and a machine learning model core **606** configured to generate output **646**. For example, output **646** may include masks **624**. The combiner **604** may be configured to combine the input audio signal **620** and the voice signatures **622** into a signal stream to be provided to the machine learning model core **606**.

(85) FIG. **6C** illustrates an example signal stream that may be concatenated from an audio signal

and voice signature(s) of target speaker(s) in the combiner **604**, according to a non-limiting embodiment of the present application. As shown in FIG. **6C**, a plurality of sequential segments may be formed as inputs to the machine learning core **606**, each including a vector representing a respective audio segment from the input audio signal appended by the voice signature(s) of one or more target speakers. A vector representing an audio segment from an input audio signal may be the output of a STFT as previously described. For example, the system may convert the audio segment into time-frequency domain vector by taking an STFT of the signal as previously described. As described above, a voice signature may also be a vector, e.g., a multi-dimensional feature vector. The STFT operation may be performed inside the voice isolation network **602** (e.g., inside combiner **604**, or inside another component in the voice isolation network, now shown), or outside the voice isolation network **602**. In the latter case, a component in the ear-worn device may convert the audio signal detected by the microphone(s) into vectors representing audio segments in the audio signal and provide the vectors (instead of audio signal) to the voice isolation network **602**. As shown in FIG. **6C**, the audio segment vectors in the inputs may be representative of sequential segments in the audio signal. For example, an audio signal may be segmented into multiple sequential segments, each converted to a vector as described above, such as seg1, seg2, seg3, etc., where each of the sequential segments may include an audio signal frame within a small time period. In some examples, an audio signal frame may have a length of 1 ms, 2 ms, 3 ms, 4 ms, or 5 ms. In other examples, the audio signal frame may have a length of greater than 30 ms (and less than one second), or any suitable time period. In a multi-speaker conversation, each of the target speakers may be associated with a distinctive voice signature. For example, vs1, vs2, vs3 may respectively represent voice signatures of target speaker 1, target speaker 2, and target speaker 3.

(86) An example of a concatenated signal stream is shown in FIG. **6C**. For example, if two target speakers are selected, the first input **632** to the machine learning core **606** may include vector seg1 representing a first audio segment appended by two voice signature feature vectors vs1 and vs2. The second input **634** to the machine learning core **606** may include vector seg2 representing a second audio segment appended by two voice signature feature vectors vs1 and vs2. The third input **636** to the machine learning model core **606** may include vector seg3 representing a third audio segment appended by two voice signature feature vectors vs1 and vs2. Additional inputs to the machine learning model core **606** may take the same, or a similar, format. Using the voice signatures in the operation of the voice isolation machine learning model enables the machine learning model, when properly trained, to isolate the speech of target speaker(s) from non-target speaker(s) in the audio signal. The training of the voice isolation machine learning model will be further described with reference to FIG. **8**.

(87) Returning to FIG. **6A**, the voice isolation machine learning model **602** may be a recurrent neural network, a convolutional neural network or any other suitable neural network. The output of the voice isolation machine learning model may include a mask **624** for isolating speeches from target speaker(s). In operation, a source separator **608** may be configured to apply the mask **624** outputted from the voice isolation machine learning model **602** to the input audio signal to provide isolated speech **610**. As shown in FIG. **6A**, isolated speech **610** may include a plurality of speech components each corresponding to a class of sounds or an individual speaker. For example, the isolated speech may include speech components of target speaker 1, target speaker 2, and non-target speaker(s) or background noise.

(88) In some embodiments, the voice isolation network may be configured to isolate speech of any suitable number of multiple target speakers with proper training datasets. For example, a voice isolation network may be configured to handle a dynamically changing number of target speakers. For example, the voice isolation network may be configured to isolate speech of up to four target speakers, including 0, 1, 2, 3, and 4 target speakers. In implementing this, the network may be configured to have the size of input and output for four target speakers, where the input and output

may contain multiple zeros where there are fewer than four target speakers. In training the network, various training datasets may be provided for various scenarios including speech of up to four distinct speakers. In such a configuration, the input to the voice isolation network (e.g., see FIG. 6C) may include a vector representing an audio segment appended by four feature vectors for four voice signatures, where some of the feature vectors may be zero when there are fewer than four target speakers. Additionally, and/or alternatively, the voice isolation network may be configured to isolate multiple target speakers by taking a combined voice signature from those of multiple target speakers. For example, the voice signatures of multiple target speakers may be combined by averaging them, where the averaged voice signature may be provided to the voice isolation network. In training the network, the voice signatures from multiple speakers may be combined in a similar manner before being provided to the voice isolation network being trained.

(89) In a non-limiting example, a mask **624** outputted by the voice isolation machine learning model **602** may include complex values. When the mask is applied to the input audio signal, the magnitude and phase of the input audio signal are modified to yield the output signal stream including the isolated speech from the target speaker(s). In some embodiments, a mask outputted by the voice isolation machine learning model may be specific to an individual target speaker or multiple target speakers. Thus, the isolated speech **610** may include signals including speeches of one or more target speaker(s). Subsequently, the signals attributable to non-target speaker(s) and/or noise may be obtained by subtracting speeches attributable to the target source(s) from the input audio signal.

(90) With continued reference to FIG. 6A, voice isolation network **600** may include additional components to further process the isolated speech to provide enhanced speech with the speech of the target speaker(s) being preferentially treated. As described above, the isolated speech **610** may include isolate speeches corresponding to different sound sources, for example, different target speakers and non-target speakers or noise. The ear-worn device may provide the isolated speech as different bands to a relative gain filter **612**, which applies different gains based on user preferences **616**. As described above, user preferences **616** may contain information about the optimal combination (or optimal weights) of various sound sources. Recombiner **614** then combines the differentially weighted frequency bands to form a combined output audio signal.

(91) Referring again to FIGS. 4A and 4B, voice isolation network **600** directs the recombined audio stream to DSP **404** or **440** for further processing. In this manner and according to one embodiment, voice isolation network **600** estimates an ideal ratio mask that separates speech signal from noise signal, applies differential gain to each of the identified speech and noise signals and combines the differentially amplified signals into one data stream.

(92) In some embodiments, voice isolation network **600** may optionally include a performance monitor **618**. Performance monitor **618** may be configured to receive output of the voice isolation machine learning model **602** to predict the performance or predict the error of the voice isolation machine learning model. These predictions can further be used as inputs in recombiner **614**, which seeks to optimize the way in which model outputs are recombined to form a final signal. Recombiner **614** takes into account both the user preferences **616** and output of performance monitor **618** to optimally recombine the outputs of isolated speech from the voice isolation machine learning model **602**.

(93) In an exemplary embodiment, performance monitor **618** receives output signal from the voice isolation machine learning model in sequential frames and determines an SNR for each frame. Performance monitor **618** then estimates an average SNR for the environment, which can be used to predict model error (since model error typically increases at more challenging input SNRs). Recombiner **614** also receives user preferences **616**. Given the user preferences **616** and the estimated SNR from the performance monitor **618**, recombiner **614** then determines a set of relative gains for the relative gain filter **612** to be applied to the isolated speech from the voice isolation machine learning model. In an exemplary embodiment, the recombiner **614** seeks to set

the gains to best match user preferences while keeping total error below a certain threshold.

(94) In some embodiments, recombiner **614** applies the gain values to the isolated speech to obtain output audio signal. In one embodiment, a plurality of gain values is communicated to recombiner **614**. Each gain values corresponds to an intermediate signal, which in turn corresponds to a sound source. Recombiner **614** multiplies each gain value to its corresponding intermediate signal and combines the results to produce output audio signal. In some embodiments, the output audio signal from the recombiner **614** may be provided to a DSP for further processing, as shown in FIGS. **4A** and **4B**.

(95) Returning to FIG. **6A**, relative gain filter **612** may receive the user's auditory preferences from user preferences **616** and apply one or more relative gains to each of the frames of isolated speech signal. In some embodiments, the gains applied to the different frequency bands in the isolated speech can be non-linear. The implementation allows different gains to be applied at the source and at per-frame level. In a non-limiting example, the relative gains may be set to cause the relative gain filter to increase the SNR of the target speaker(s). For example, a higher gain may be applied to the speech component attributable to the target speaker(s) and a lower gain may be applied to the speech component attributable to non-target speaker(s) or background noise. Alternatively, the relative gains may be set to cause the relative gain filter to set the volume of the speech of the target speaker(s) to be at a desirable level. For example, the desirable level for a target speaker may be configured in the user preferences **616**. In other variations, the volume of each target speaker may be equalized among the multiple target speakers. In some embodiments, the volume of each target speaker may be adjusted independently depending on how far the speaker is from the wearer of the ear-worn device. Adjusting the volumes of the target speaker(s) may include applying a higher gain to the speaker sitting farther away so that the volume of the voice at the receiver(s) of the ear-worn device is the same as that of the voice of the person sitting closer. In some embodiments, information indicative of the distance from a speaker to the wearer may be provided to the ear-worn device via an application on the external electronic device associated with the wearer. In some embodiments, the ear-worn device may adjust the volumes of speakers based on the control of the wearer (e.g., via an application on the external electronic device).

(96) In some embodiments, the relative gains may be set to cause the relative gain filter to attenuate speech of non-target speaker(s) and/or the background noise. Additionally, and/or alternatively, the ear-worn device may receive an indication (e.g., set on the phone associated with the wearer of the ear-worn device, or set on the ear-worn device with a user selection) to suppress the wearer's own speech. In such configuration, the wearer him/herself may be designated as a non-target speaker whose speech should be isolated by the voice isolation network in the manner described herein for target speakers, but whose speech should be de-emphasized or suppressed. Once the wearer's own speech is isolated, the ear-worn device may set the relative gains to attenuate/suppress the wearer's speech. As the result, the output signal may be enhanced to include only speech of target speaker(s) other than the wearer of the ear-worn device. It is appreciated that the voice isolation network **600** may be implemented in configuration as described in FIG. **4B**, in which the voice isolation network may be activated or deactivated by a controller (e.g., **430**) as described above.

(97) FIG. **7A** is a flowchart of an example method **700** of operation of an ear-worn device configured to selectively isolate speech from a target speaker within a multi-speaker environment, according to a non-limiting embodiment of the present application. In some embodiments, method **700** may be implemented by a processor in an ear-worn device such as **108** (in FIGS. **1** and **2**), **302** (in FIG. **3**), **400** (in FIG. **4A**), **470** (in FIG. **4B**), **500** (in FIG. **5A**) or **545** (in FIG. **5B**).

(98) Method **700** may implement any of the operations in various embodiments described above. For example, method **700** may include detecting an audio signal with a microphone of an ear-worn device at act **702**, providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device at act **704**; and isolating, with the processor of the ear-worn device, speech of target speaker(s) with a machine learning model using voice signature(s) of the

target speaker(s), at act **708**. In some embodiments, the machine learning model may be a voice isolation machine learning model in a voice isolation network, such as voice isolation network described above in FIGS. **6A-6B**. The machine learning model **602** may receive two inputs: the audio signal detected by the microphone of the ear-worn device and voice signature(s) of the target speaker(s). Voice signature of a speaker, as previously described, may be a unique representation of the speaker that differentiates the voice of the speaker from those of other speakers. Using the voice signature of target speaker(s) as input, the machine learning model may be operated to isolate the speech from the target speaker(s) in the audio signal. The isolated speech may include a component of the audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers.

(99) In some embodiments, the voice signature(s) of the target speaker(s) may be obtained from another machine learning model trained to discriminate between voices of speakers. With reference to FIG. **7A**, method **700** may optionally include receiving voice signature(s) of target speaker(s) at act **706**. For example, as shown in FIG. **3**, the ear-worn device **302** may wirelessly receive the voice signature(s) of target speaker(s) from an electronic device **304**, such as a phone associated with the wearer of the ear-worn device. In some embodiments, the ear-worn device **302** may be configured to send a segment of an audio signal detected from the microphone of the ear-worn device to electronic device **304**, which may process the audio signal to extract voice signature(s) of the target speaker(s). Accordingly, act **706** may further include sending a segment of an audio signal to an external electronic device in the manner as described in embodiments of FIG. **3** before receiving the voice signature(s) of the target speaker(s).

(100) FIG. **7B** illustrates a flowchart of an example method **750** as a variation of the example method in FIG. **7A** of operation of an ear-worn device configured to selectively isolate speech from a target speaker within a multi-speaker environment, according to a non-limiting embodiment of the present application. Similar to method **700** in FIG. **7A**, method **750** may be implemented by a processor in an ear-worn device such as **108** (in FIGS. **1** and **2**), **302** (in FIG. **3**), **400** (in FIG. **4A**), **470** (in FIG. **4B**), **500** (in FIG. **5A**) or **545** (in FIG. **5B**). In some embodiments, method **750** may include detecting audio signal with a microphone of an ear-worn device at **752** and providing the audio signal to the processor of the ear-worn device at act **754**, where acts **752** and **754** may be respectively performed in a similar manner as acts **702** and **704** in method **700**.

(101) Additionally, and/or alternatively, method **750** may include increasing SNR of the target speaker(s) with a machine learning model using voice signature(s) of the target speaker(s), at act **758**. Similar to method **700**, the machine learning model may be a voice isolation machine learning model in a voice isolation network, such as voice isolation network described above in FIGS. **6A-6B**. For example, the machine learning model may receive two inputs: the audio signal detected by the microphone of the ear-worn device and voice signature(s) of the target speaker(s), and output isolated speech from the target speaker(s). Further, act **758** may include one or more operations that may be performed in the voice isolation network as described in FIG. **6B**. For example, recombiner **614** may be operated to output audio signal including improved SNR of the target speaker(s).

(102) At act **756**, similar to act **706** of method **700**, the voice signature of target speaker(s) to be used with the machine learning model may be received (e.g., wirelessly) from an external electronic device. Further, act **756** may include sending a segment of an audio signal to an external electronic device in the manner as described in embodiments of FIG. **3** before receiving the voice signature(s) of the target speaker(s).

(103) In both acts **706** and **756**, in some embodiments, the voice signatures of multiple speakers are stored in the external electronic device, and thus, the voice signature(s) of the target speaker(s) are received from the external electronic device. In other embodiments, voice signatures of multiple speakers may be stored in the ear-worn device. As such, the identifiers that identify the target speaker(s) are received by the ear-worn device, which may use the identifiers to retrieve the corresponding voice signatures of the target speaker(s) from the local storage of the ear-worn

device. In some other embodiments, rather than receiving voice signature(s) from an external device, acts **706/756** may include generating the voice signature(s) by the ear-worn device itself using a voice signature network that is also residing in the ear-worn device. In some other embodiments, voice signature(s) representing default target speakers may be pre-stored on the ear-worn device. In such configuration, acts **708/758** may include retrieving the voice signature(s) from a memory location of the ear-worn device and providing the voice signature(s) to the machine learning model.

(104) In some embodiments, the target speaker(s) may be maintained the same during a conversation session, e.g., a meeting, a diner, in which the participants in the conversation do not change. In such case, the voice signature(s) of the target speaker(s) may be selected at the beginning of the conversation and remain constant throughout the conversation session. Thus, acts **708/758** (of method **700/750**) may include providing the voice signature as a constant input to the machine learning model during processing of the audio signal. Alternatively, the model may be configured such that it continues to target those voice signatures previously provided, until it receives a new voice signature or an instruction indicating no voice signature. In such configurations, the machine learning model may be run in an efficient manner, obviating the need to keep passing large inputs to the network. In some embodiments, an indication of no voice signature may include a voice signature being set to a default value (e.g., all zeros in a voice signature). In some embodiments, an indication of no voice signature may cause the machine learning model to operate to isolate all voices.

(105) In some embodiments, the target speaker(s) may change during a conversation. This may happen when the participants in the conversation change (e.g., a speaker left the conversation, or a new speaker joined), or when the wearer of the ear-worn device decides to listen to a different target speaker. In some embodiments, the wearer of the ear-worn device may be a user of his/her phone during a conversation, as shown in FIG. 1. Thus, the user may change target speaker(s) by making a user selection on the phone. In other embodiments, the ear-worn device may periodically transmit an audio signal segment detected by the microphone(s) to the external electronic device, which generates and sends updated voice signature(s) of target speaker(s) to the ear-worn device. At such time when the target speakers are updated, acts **706/756** (of method **700/750**) may be triggered to receive updated voice signature(s) of the target speaker(s). Consequently, acts **708/758** (of method **700/750**) may include providing a second voice signature as input to the machine learning model in place of the previous voice signature during processing of the audio signal.

(106) It is appreciated that each of methods **700** and **750** may include one or more additional acts to implement one or more operations described above to enhance the audio signal. For example, by the DSP (**404** of FIG. 4A or **440** of FIG. 4B), the ear-worn device may play out only the target speaker's speech after isolating the component of the audio signal representing the target speaker's speech. In implementing this, methods **700** or **750** may amplify the component of the audio signal representing the target speaker's speech and/or apply a lower gain to or not amplify the speech component attributable to the non-target speaker(s) or background noise. Alternatively, and/or additionally, method **700** or **750** may include executing the recombiner **614** of FIG. 6A to suppress the component of the audio signal representing a non-target speaker's speech after isolating the component of the audio signal representing the target speaker's speech.

(107) FIG. 8 is a block diagram illustrating at **800** the training and deploying of a voice isolation machine learning model for isolating speech from a target speaker, according to a non-limiting embodiment of the present application. In some embodiments, a voice isolation machine learning model training system **802** may be configured to use training dataset **804** to generate a trained voice isolation machine learning model **806**. The trained voice isolation machine learning model **806** may be deployed as the voice isolation machine learning model **602** in the voice isolation network **600**, for example. The train voice isolation machine learning model **806** may also be implemented in voice isolation network described above, such as **403** (in FIG. 4A), **450** (in FIG. 4B), **540** (in FIG.

5A), 570 and 580 (in FIG. 5B). In some embodiments, the voice isolation machine learning model 806 may include a neural network comprising a plurality of layers, each having multiple weights. It is appreciated that the voice isolation machine learning model 806 may be any suitable model configured in a suitable manner. In a non-limiting example, the machine learning model 806 may be a recurrent neural network and may be a long short term memory (LSTM) network. In some examples, the machine learning model 806 may include five LSTM layers, each layer having a number of units, e.g., 1024 or other suitable number of units. The number of weights in the machine learning model 806 may be as few as thousands to as large as tens of millions. Training system 802 may use training dataset 804 to train the weights in the machine learning model 806. (108) In some embodiments, training dataset 804 may include clips of clean speech 810 and noisy speech 812 for a plurality of speakers. For example, for each of the plurality of speakers, the training dataset 804 may include a plurality of pairs of clips containing speech from the speaker. Each pair of clips may include a clip containing clean speech and another clip containing the same clean speech with interfering noise added (thus noisy speech). The interfering noise may include randomly selected background noise and also interfering contemporaneous speech of a non-target speaker or other speakers. The data can also be augmented by adding reverberation, in some examples. Thus, training system 802 receives the plurality of pairs of clips for the plurality of speakers as input. Additionally, training system 802 may also receive voice signatures of the plurality of speakers as a third input.

(109) In training the voice isolation machine learning model 806, a pre-trained voice isolation machine learning model is initialized and contains initial weights. The training system then provides the noisy speech and the voice signatures 824 to the pre-trained voice isolation machine learning model to generate an output for each data point in each of a plurality of iterations in an optimization process. For example, each data point in an iteration may be a small segment of the noisy speech (e.g., a chunk of one second) of a speaker combined with the voice signature for the speaker. The clips of clean speech may be used as ground truth data. In some embodiments, the segment of the noisy speech and the voice signature of the speaker may be concatenated in the same manner as described in FIG. 6C as used in executing the voice isolation network (or voice isolation machine learning model). In some embodiments, the output of the voice isolation machine learning model for each data point may be an estimated complex mask for the noisy clip. A loss function is calculated based on the difference (e.g., measured by mean-squared error) between the ground truth complex mask and the estimated complex mask. The objective of the training is to minimize such loss function through multiple iterations in the optimization process. Any suitable algorithm, such as gradient descent, may be used in the optimization process. Once the training is completed, the weights in the voice isolation machine learning are trained.

(110) Voice signatures 824 provided to the training system 802 may be pre-stored or generated concurrently with the training. In some embodiments, voice signatures of plurality of speakers may be extracted from audio signals containing speeches of the speakers using a voice signature network, which will be described in detail with reference to FIGS. 12-13. The generated voice signatures may be pre-stored, e.g., together with the training system. In other embodiments, training dataset 804 may include additional clips of clean speech 822 for the plurality of speakers, which are provided to a trained voice signature machine learning model 808. The trained voice signature machine learning model 808 is then executed to generate the voice signature for each additional clip. As such, the training dataset 804 includes a triplet of clips for each speaker, including two clips of clean speech and a clip of noisy speech. In this configuration, one clip of clean speech is used as ground truth, the other clip of clean speech is used to generate a voice signature. The generated voice signature and the clip of noisy speech are then provided to the training system 802.

(111) The trained voice isolation machine learning model 806 may be deployed (executed) in the voice isolation network described in various embodiments above. The trained voice isolation

machine learning model may take any new noisy speech **814** along with the voice signature(s) of the target speaker(s) **816** and generate results **818** (e.g., complex masks as described above). The new noisy speech and the voice signature(s) of target speaker(s) may be combined in a similar manner as described in FIG. 6C, and such manner of combination will therefore not be repeated herein.

(112) FIG. 9 illustrates a block diagram of a system-on-chip (SOC) package that may be implemented in an ear-worn device, according to a non-limiting embodiment of the present application. In some embodiments, SOC package **902** may implement various operations in an ear-worn device, such as **108** (in FIGS. 1 and 2), **302** (in FIG. 3), **400** (in FIG. 4A), **470** (in FIG. 4B), or a circuitry of an ear-worn device such as **500** (in FIG. 5A), or **545** (in FIG. 5B). In various embodiments, SOC **902** includes one or more Central Processing Unit (CPU) cores **920**, an Input/Output (I/O) interface **940**, and a memory controller **942**. Various components of the SOC package **902** may be optionally coupled to an interconnect or bus such as discussed herein with reference to the other figures. Also, the SOC package **702** may include components such as those discussed with reference to the ear-worn device described in FIGS. 1-8. Further, each component of the SOC package **920** may include one or more other components of the ear-worn device, e.g., as discussed with reference to FIGS. 4A-6B. In one embodiment, SOC package **902** (and its components) is provided on one or more Integrated Circuit (IC) die, e.g., which are packaged into a single semiconductor device. The single semiconductor device may be configured to be used as an ear-worn device, an amplification system or a hearing device to be used in the human ear canal.

(113) As illustrated in FIG. 9, SOC package **902** is coupled to a memory **960** via the memory controller **942**. In an embodiment, the memory **960** (or a portion of it) can be integrated on the SOC package **902**. The I/O interface **940** may be coupled to one or more I/O devices **970**, e.g., via an interconnect and/or bus such as discussed herein. I/O device(s) **970** may include interfaces to communicate with SOC **902**. In an exemplary embodiment, I/O interface **940** communicates wirelessly with I/O device **970**. SOC package **902** may comprise hardware, software and logic to implement, for example, the various components or methods described in FIGS. 1-8. The implementation may be communicated with an auxiliary device, e.g., I/O device **970**. I/O device **970** may comprise additional communication capabilities, e.g., cellular, Bluetooth, WiFi or other protocols, to access any component in the ear-worn device, for example, to configure the voice isolation network.

(114) FIG. 10 is a block diagram illustrating a portion of a circuitry configuration of an electronic device operable to extract voice signature of target speaker(s) to an ear-worn device, according to a non-limiting embodiment of the present application. In some embodiments, the wearer of an ear-worn device may be a user of his/her phone during a conversation, as shown in FIG. 1, and use his/her phone to perform various operations in association with voice processing on the ear-worn device. For example, circuitry **1000** may be implemented in electronic device **110** (in FIGS. 1 and 2), and **304** (in FIG. 3) to provide the voice signature(s) for target speaker(s) **1018** and transmit (e.g., wired or wirelessly) the voice signature(s) to the ear-worn device (see FIG. 3). Circuitry **1000** may include various components, either hardware- or software-implemented. In some embodiments, circuitry **1000** may include a voice signature network **1002** described above. The voice signature network **1002** may be configured to receive an audio signal **1014** as input and extract as output voice signature of the speaker(s) **1016** whose speech is included in the audio signal. In some embodiments, the voice signature network may include a voice signature machine learning model that can be trained. When properly trained, the voice signature machine learning model may be executed to extract from input audio signal distinctive voice signatures for different speakers and extract the same voice signature for the same speaker. A voice signature may be in the form previously described, e.g., in a multi-dimensional feature vector. The training of the voice signature machine learning model is described with reference to FIG. 13.

(115) With further reference to FIG. 10, circuitry **1000** may include a target speaker selector **1004**,

a user interface **1006** coupled to the target speaker selector **1004**. Circuitry **1000** may further include a speaker registry **1008** to store a list of known speakers to the wearer of the ear-worn device, and storage **1010** to store voice signatures of speakers. In some embodiments, the voice signatures **1010** of known speakers may be stored in association with registry **1008** of known speakers, where each entry of the registry may correspond to a respective individual speaker. As shown in FIG. **10**, registry **1008** and voice signatures **1010** may be stored on the electronic device. In other embodiments, the voice signatures may be optionally and/or additionally stored on the ear-worn device associated with the electronic device. In some embodiments, circuitry **1000** may include a voice signature collector **1012**, which is configured to collect the voice signatures **1010**. The collection of voice signatures will be further described with reference to FIG. **12**.

(116) As shown in FIG. **10**, target speaker selector **1004** may be configured to select target speaker(s) for the user and transmit the voice signature(s) of selected target speaker(s) to the user's ear-worn device. The operations of the target speaker selector **1004** are further explained with examples in FIGS. **11A-11F**. FIGS. **11A**, **11C**, and **11E** illustrate examples of graphical user interface that may be implemented in an electronic device, according to some non-limiting embodiments of the present application. FIGS. **11B**, **11D**, and **11F** illustrate block diagrams of example processes respectively for implementing the example graphical user interfaces shown in FIGS. **11A**, **11C**, and **11E**.

(117) With reference to FIG. **11A**, an example display **1100** may be implemented in user interface **1006** of FIG. **10**. The display of the user interface **1100** may include a list **1102** of known speakers to user (e.g., the wearer of the ear-worn device). The known speakers may be registered in the speaker register (e.g., **1008** in FIG. **10**). The user may view the listed speakers and confirm who is present in a conversation and/or whom the user would like to listen to. This may be implemented by the user providing a user selection of target speaker(s) from the list of speakers in the registry. As shown in the display **1100**, each speaker name may be displayed with a check box **1106** for the user to select/unselect. Once the user has made the selection, the user may click a “Send” button **1104** to send the voice signature(s) of selected target speaker(s) to the ear-worn device. In some embodiments, the user may not need to click a “Send” button. Instead, once the target speakers are selected/updated (automatically, or by the user), the electronic device may dynamically send the voice signature(s) of the updated target speaker(s) to the ear-worn device.

(118) FIG. **11B** illustrates an example process **1120** for implementing the example graphical user interface of FIG. **11A**. In some embodiments, method **1120** may be implemented in an electronic device, e.g., **110** (in FIGS. **1** and **2**), **304** (in FIG. **3**). For example, method **1120** may be implemented in the target speaker selector **1104** of FIG. **10**. In some embodiments, method **1120** may include displaying identities of known speaker(s) in a registry at act **1122**. For example, as shown in FIG. **11A**, the user interface may display a list **1102** of known speakers in the registry. Method **1120** may further include receiving user selection identifying the target speaker(s) at act **1124**. Although it is shown in FIG. **11A** that the user selection may be click(s) **1106**, the user selection may also take other forms, such as drop-down menu or other suitable widgets.

(119) In response to receiving the user section at act **1124**, method **1120** may proceed to determine whether there is at least one target speaker selected, at act **1126**. In response to determining that at least one target speaker is selected, method **1120** may proceed to transmit the voice signature(s) associated with the selected speaker(s) to the ear-worn device, at act **1128**. For example, the method may enable the user to click “Send” button **1104** of FIG. **11A** to transmit the voice signature(s). In another example, the method may dynamically transmit the voice signature(s) once the selected speaker(s) are updated, without requiring the user to click a “Send” button. In response to determining that no target speaker is selected, or no user selection is received, method **1120** may stop. In some embodiments, in response to determining that no target speaker is selected, or no user selection is received, method **1120** may disable the “Send” button **1104** of FIG. **11A** (e.g., the “Send” button may be grayed out), in which case, the electronic device will not transmit any voice

signature to the ear-worn device.

(120) With reference to FIG. **11C**, an example display **1140** may be implemented in user interface **1006** of FIG. **10**. The display **1140** may include a list **1102** of one or more identified speakers. In some embodiments, the electronic device may receive audio signal detected by the microphone of the ear-worn device (or the electronic device itself) in a conversation to identify one or more speakers in the conversation. A speaker may be identified when an extracted voice signature from the audio signal is matched to a voice signature associated with a registry of known speakers. Extracting voice signature(s) from an audio signal will be further described in FIG. **11D**. In some embodiments, the electronic device may transmit the voice signature(s) of the identified one or more speakers to the ear-worn device, assuming, by default, any known speakers in the registry who are also present in a conversation (thus, the voice signature is matched, and the speaker is identified) are target speakers to whom the wearer of the ear-worn device prefers to listen. Additionally, in the user interface shown in **1140**, the user may be prompted to select, at **1146**, a subset of the identified speakers to further confirm the target speaker(s). Then, the user may proceed to transmit the voice signatures of the selected target speaker(s), e.g., by clicking the “Send” button **1144**.

(121) Additionally, and/or alternatively, the list **1142** may include one or more un-identified speakers whose voice signature(s) are not matched to any of the voice signatures of known speakers in the registry. An un-identified speaker may be a new speaker whom the wearer of the ear-worn device has never spoken with before. In some embodiments, the user interface may allow the user to add a new un-identified speaker to the registry, e.g., by clicking “Update registry” button **1150**. Once the user selects an un-identified speaker to be added to the registry, the un-identified speaker may be added to the registry. The newly added speaker may also be automatically selected as a target speaker.

(122) Additionally, and/or alternatively, the user interface **1140** may include a user selection **1148** that, when selected, enables the user to suppress his/her own voice. Such user selection may be an indication for suppressing the wearer's own voice, where the indication can be transmitted from the electronic device to the ear-worn device. Alternatively, the ear-worn device may include a user interface (e.g., by user clicking one or more buttons or a combination of buttons) to configure the ear-worn device to suppress the wearer's own voice. In an alternative embodiment, the ear-worn device may be configured, by default, to suppress the wearer's own voice. The techniques for suppressing the wearer's own voice are previously described with reference to FIG. **6A**, and the description of those techniques will not be repeated herein.

(123) FIG. **11D** illustrates an example process **1160** for implementing the example graphical user interface of FIG. **11C**. In some embodiments, method **1160** may be implemented in an electronic device, e.g., **110** (in FIGS. **1** and **2**), **304** (in FIG. **3**). For example, method **1160** may be implemented in the target speaker selector **1104** of FIG. **10**. In some embodiments, method **1160** may include receiving input speech signal at act **1162**. As previously described, the electronic device may receive the input speech signal from an ear-worn device. For example, the input speech signal may be detected by the microphone(s) of the ear-worn device and transmitted wirelessly to the electronic device. Alternatively, or additionally, the electronic device may receive the input speech signal from its own microphone(s), as described in FIG. **3** (showing the audio signal to be processed in the electronic device may come from either the ear-worn device and/or the electronic device itself). As described above, the input speech signal may include the speech from one or more target speakers. The input speech signal may also include the speech from one or more non-target speakers. In some examples, the input speech signal may be detected in a beginning of a multi-speaker conversation, during which period every speaker (or every target speaker) has spoken at least once.

(124) With further reference to FIG. **11D**, method **1160** may include extracting voice signature(s) of speakers from the input speech signal using a voice signature network at act **1164**. The voice

signature network may be configured as described previously as in **1002** of FIG. **10**, for example. Using the voice signature network, act **1164** may extract the voices of the speakers in the conversation based on the input speech signal. Method **1160** may further match the extracted voice signatures, at act **1166**, with the voice signatures (e.g., **1010** of FIG. **10**) associated with the known speakers in the registry (e.g., **1008** of FIG. **10**). If an extracted voice signature is matched to a voice signature associated with a known speaker, method **1160** may identify that known speaker. Method **1160** may further display a respective entry of the identified speaker in the registry at act **1168**. For example, list **1142** in the user interface **1140** displays the names of the speakers in the registry whose voice signatures respectively match an extracted voice signature from the input speech signal.

(125) In some embodiments, method **1160** may handle an unidentified speaker. An example of an “un-identified” entry is shown in list **1142** of FIG. **11C**. Method **1160** may determine whether an extracted voice signatures from the input speech signal is not matched to any of the voice signatures of the known speakers in the registry at **1170**. In response to determining an unmatched voice signature, method **1160** may proceed to enable the user of the electronic device to update the registry of speakers at act **1172**, for example, by clicking “Update registry” button **1150** of FIG. **11C**. Updating the registry will further be described in FIGS. **11E** and **11F**.

(126) With continued reference to FIG. **11D**, in response to determining no unmatched voice signatures, method **1160** may optionally receive user selection identifying target speaker(s) from the matched speakers at act **1174**. Before receiving user selection identifying the target speaker(s), act **1174** may display the user with an option to select the target speaker. For example, a list of selection boxes **1146** of FIG. **11C** may be displayed aside the list of identified speakers **1142**. Each of the selection boxes **1146** of FIG. **11C** may be clickable to allow the user to click/unclick a corresponding speaker in the list. Thus, the selected target speaker may be a subset of the identified speakers described above. For example, as shown in FIG. **11C**, the user selection may include selecting “John Rogers,” “Sarah Smith,” and “Clark Hamm.”

(127) Returning to FIG. **11D**, in response to receiving the user section at act **1174**, method **1170** may proceed to determine whether there is at least one target speaker selected, at act **1176**. In response to determining that at least one target speaker is selected, method **1160** may proceed to transmit the voice signature(s) associated with the selected target speaker(s) to the ear-worn device, at act **1178**. For example, method **1160** may enable the user to click “Send” button **1144** of FIG. **11C** to transmit the voice signature(s) to the ear-worn device. In another example, the method may dynamically transmit the voice signature(s) to the ear-worn device upon matching the extracted voice signature(s) to the registry of known speakers at act **1166**, without waiting for any user selection. In response to determining that no target speaker is selected, method **1160** may stop. In some embodiments, in response to determining that no target speaker is selected, or no user selection is received, method **1160** may disable the “Send” button **1144** of FIG. **11C** (e.g., the “Send” button may be grayed out), in which case, the electronic device will not transmit any voice signature to the ear-worn device.

(128) As previously described, the ear-worn device may additionally store voice signature(s) of one or more speakers. For example, see voice signature storage **316** of FIG. **3**. In some embodiments, the voice signature(s) stored in the ear-worn device may be voice signature(s) of default target speaker(s). Alternatively, the voice signature(s) stored in the ear-worn device may be voice signature(s) of all known speakers to the wearer of the ear-worn device. Alternatively, the voice signature(s) stored in the ear-worn device may be associated with speakers with whom the wearer of the ear-worn device have frequently spoken, e.g., the voice signature(s) may be updated periodically or dynamically. It is appreciated that the voice signature(s) stored on the ear-worn device may be uploaded/updated by a user interface (e.g., click of buttons or a combination of buttons, or voice commands etc.) or via an electronic device (e.g., a phone).

(129) In case voice signatures of some speakers are stored on the ear-worn device, acts **1128** of

FIG. 11B and 1178 of FIG. 11D may instead transmit the identifiers of selected target speaker(s) rather than the voice signature(s) themselves. For example, the identifiers of selected target speaker(s) may include the speaker's names such as the names shown in list 1102 of FIG. 11A and 1142 of FIG. 11C. In other examples, the identifier(s) of selected speaker(s) may each include a fixed-length string, a multiple-digit code, or any other suitable identifiers. The ear-worn device may use the identifier(s) to retrieve the associated voice signature(s) of the selected target speaker(s).

(130) It is appreciated that the electronic device may be configured to enable to the user to edit the registry of known speakers at any time. In some embodiments, the registry may be edited to store only default speakers. In other embodiments, the registry may be edited to store known speakers to the wearer of the ear-worn device. In some embodiments, the registry on the electronic device may correspond to the voice signatures stored on the ear-worn device. In other embodiments, the registry on the electronic device may be independent of the voice signatures stored on the ear-worn device.

(131) In FIG. 11E, updating the registry is further illustrated. In some embodiments, the user interface 1180 is triggered by a click of “Update registry” button 1150 of FIG. 11C. In the scenario previously described in FIG. 11C, the voice signature of an un-identified speaker may be extracted from the input speech signal but not matched to any voice signature of known speakers in the registry. In FIG. 11E, the user interface 1180 prompts the user to enter the name of the new speaker via a widget, such as an input box 1182. The user interface 1180 may receive a user entry of the name of the new identify. After entering the name, the user may click an “Update” button 1184 to update the registry with the new entered name.

(132) FIG. 11F illustrates an example process 1190 for implementing the example graphical user interface of FIG. 11E, according to a non-limiting embodiment of the present application. In some embodiments, method 1190 may be implemented in an electronic device, e.g., 110 (in FIGS. 1 and 2), 304 (in FIG. 3). For example, method 1190 may be implemented in the target speaker selector 1104 of FIG. 10. In some embodiments, method 1190 may include receiving a user input for a new speaker identity at act 1192. For example, the new speaker identity may be entered by the user in an input box (e.g., 1182 of FIG. 11E) in a user interface. Method 1190 may further include adding the new speaker identity to the registry at act 1194, and storing the voice signature of the new speaker associated with the registry at act 1196. Accordingly, with reference to FIG. 10, the electronic device may store the unmatched voice signature in the voice signatures storage (e.g., 1010 of FIG. 10), update the registry (e.g., 1008 of FIG. 10) with a new entry associated with the new name, and associate the new entry with the unmatched voice signature.

(133) FIG. 12 illustrates a block diagram of an example process 1200 for collecting a voice signature of a speaker, according to a non-limiting embodiment of the present application. As described above, the voice signatures of speakers may be pre-collected and stored on an electronic device for use with selecting the target speakers, as described in FIG. 10 and various examples in FIGS. 11A-11F. In some embodiments, method 1200 may be implemented in the voice signature collector 1012 of FIG. 10. Collecting the voice signatures of a plurality of speakers may include collecting a respective audio segment for each of the plurality of speakers; generating a respective voice signature for each of the plurality of speakers using a neural network over the respective audio segment; and registering in a registry the voice signatures of the plurality of speakers with the plurality of speakers in the contact list. In some embodiments, in collecting voice signature of a speaker, method 1200 may include receiving an audio segment (e.g., a segment of audio signal) including the speech of the speaker at act 1204. The audio segment may be recorded in a conversation including the speaker. Alternatively, and/or additionally, method 1200 may include outputting a prompt to the user at act 1202. For example, method 1200 may display a prompt (e.g., a script) and record the audio segment in respond to the speaker reading the prompt.

(134) Method 1200 may further process the audio segment with a voice signature network to

extract a voice signature for the speaker at act **1206**. In some embodiments, the voice signature network may be implemented as **1002** of FIG. **10** for execution on an electronic device. Method **1200** may further store the extracted voice signature for the speaker and register the extracted voice signature in association with the speaker in a registry.

(135) Although method **1200** may be implemented in an external electronic device, such as a phone, in other embodiments, method **1200** may be performed on a server (e.g., **306** of FIG. **3**), or any other device. In such configuration, act **1204** may include receiving the audio segment from an electronic device (e.g., a phone capable of recording), which records the audio segment from the speaker. Act **1208** may include transmitting the extracted voice signature to the electronic device for registering with the registry.

(136) FIG. **13** is a block diagram illustrating training and deploying **1300** of a voice signature machine learning model for extracting voice signature from speech data, according to a non-limiting embodiment of the present application. In some embodiments, a voice signature machine learning model training system **1302** may be configured to use training dataset **1304** to generate a trained voice signature machine learning model **1306**. The trained voice signature machine learning model **1306** may be deployed in voice signature network described above, such as **1002** of FIG. **10**. The trained voice signature machine learning model may also be used in training the voice isolation network, as described in FIG. **8** (see **808** of FIG. **8**). In other embodiments, the trained voice signature machine learning model may also be used to implement the voice signature network for collecting voice signatures, as described in FIG. **12**.

(137) In some embodiments, the voice signature machine learning model **1306** may include a neural network comprising a plurality of layers, each having multiple weights. Training system **1302** may use training dataset **1304** to train the weights in the machine learning model **1306**. In some embodiments, the training system **1302** may train the machine learning model **1306** using a contrastive learning method. The training dataset **1304** may include pairs of audio clips including speech collected from a plurality of speakers. For example, multiple clips are collected from various different speakers, with each clip being labelled with a unique speaker ID. The clips may be organized into positive **1310** and negative pairs **1312**, where positive pairs denote pairs of clips belonging to the same speaker while negative pairs denote clips belonging to different speakers. In some embodiments, the training dataset **1304** may contain clean speech data (without noise). The training data may further be augmented with added noise. For example, audio data in the training dataset may be augmented by mixing in background audio for a small portion of the clips and applying room impulse responses to the speech to add reverberation.

(138) In training the machine learning model **1306**, a pre-trained voice signature machine learning model is initialized and contains initial weights. The training system then provides the training dataset including the positive and negative clip pairs to the pre-trained voice signature machine learning model to generate an output for each data point in each of a plurality of iterations in an optimization process. For example, each data point in an iteration may be a small segment of the clip pairs (e.g., in chunks of 1 second each). The training system **1302** may provide the clip pairs through the voice signature machine learning model **1306** and output pairs of embeddings. The optimization process may be configured in such a way that embeddings corresponding to clips from the same speaker are made as similar as possible, while embeddings corresponding to clips from different speakers are optimized to be as different as possible.

(139) In some embodiments, a contrastive loss function is applied directly to the output (embeddings) corresponding to input audio clips. In another embodiment, a contrastive loss function is applied between each embedding and the centroid of the corresponding cluster of embeddings in the latent space. The similarity may be quantified using cosine similarity and the loss function may be configured such that the optimization maximizes the cosine similarity for positive pairs and minimizes the cosine similarity for negative pairs. In some embodiments, the embedding (a 256-dimensional vector) may be obtained by averaging the outputs corresponding to

each chunk (e.g., 1 second).

(140) The trained voice signature machine learning model **1306** may be deployed (executed) in the voice signature network described in various embodiments above. The trained voice signature machine learning model may take any new audio speech signal **1314** and extract voice signature(s) of one or more speaker(s) **1316**.

(141) FIG. **14** illustrates an example of a computing system **1400** that may be implemented in an electronic device to implement various embodiments described in the present application. In some embodiments, system **1400** may implement operations described in various embodiments with reference to FIGS. **1-3** and **10-13** on an electronic device, such as **110** (in FIGS. **1** and **2**) or **304** (in FIG. **3**). In some embodiments, the system **1400** includes one or more processors **1402** and one or more graphics processors **1408**, and may be a single processor desktop system, a multiprocessor workstation system, or a server system having a large number of processors **1402** or processor cores **1407**. In one embodiment, the system **1400** is a processing platform incorporated within a system-on-a-chip (SoC or SOC) integrated circuit for use in mobile, handheld, or embedded devices.

(142) An embodiment of system **1400** can include or be incorporated within a server-based smart-device platform or an online server with access to the internet. In some embodiments system **1400** is a mobile phone, smart phone, tablet computing device or mobile Internet device. Data processing system **1400** can also include, couple with, or be integrated within a wearable device, such as a smart watch wearable device, smart eyewear device (e.g., face-worn glasses), augmented reality device, or virtual reality device. In some embodiments, data processing system **1400** is a television or set top box device having one or more processors **1402** and a graphical interface generated by one or more graphics processors **1408**.

(143) In some embodiments, the one or more processors **1402** each include one or more processor cores **1407** to process instructions which, when executed, perform operations for system and user software. In some embodiments, each of the one or more processor cores **1407** is configured to process a specific instruction set **1409**. In some embodiments, instruction set **1409** may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). Multiple processor cores **1407** may each process a different instruction set **1409**, which may include instructions to facilitate the emulation of other instruction sets. Processor core **1407** may also include other processing devices, such as a DSP.

(144) In some embodiments, the processor **1402** includes cache memory **1404**. Depending on the architecture, the processor **1402** can have a single internal cache or multiple levels of internal cache. In some embodiments, the cache memory is shared among various components of the processor **1402**. In some embodiments, the processor **1402** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **1407** using known cache coherency techniques. A register file **1406** is additionally included in processor **1402** which may include different types of registers for storing different types of data (e.g., integer registers, floating point registers, status registers, and an instruction pointer register). Some registers may be general-purpose registers, while other registers may be specific to the design of the processor **1402**.

(145) In some embodiments, processor **1402** is coupled to a processor bus **1410** to transmit communication signals such as address, data, or control signals between processor **1402** and other components in system **1400**. In one embodiment the system **1400** uses an exemplary 'hub' system architecture, including a memory controller hub **1416** and an Input Output (I/O) controller hub **1430**. A memory controller hub **1416** facilitates communication between a memory device and other components of system **1400**, while an I/O Controller Hub (ICH) **1430** provides connections to I/O devices via a local I/O bus. In one embodiment, the logic of the memory controller hub **1416** is integrated within the processor.

(146) Memory device **1420** can be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. In one embodiment the memory device **1420** can operate as system memory for the system **1400**, to store data **1422** and instructions **1421** for use when the one or more processors **1402** executes an application or process. Memory controller hub **1416** also couples with an optional external graphics processor **1412**, which may communicate with the one or more graphics processors **1408** in processors **1402** to perform graphics and media operations.

(147) In some embodiments, ICH **1430** enables peripherals to connect to memory device **1420** and processor **1402** via a high-speed I/O bus. The I/O peripherals include, but are not limited to, an audio controller **1446**, a firmware interface **1428**, a wireless transceiver **1426** (e.g., Wi-Fi, Bluetooth), a data storage device **1424** (e.g., hard disk drive, flash memory, etc.), and a legacy I/O controller **1440** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to the system. One or more Universal Serial Bus (USB) controllers **1442** connect input devices, such as keyboard and mouse **1444** combinations. A network controller **1434** may also couple to ICH **1430**. In some embodiments, a high-performance network controller (not shown) couples to processor bus **1410**. It will be appreciated that the system **1400** shown is exemplary and not limiting, as other types of data processing systems that are differently configured may also be used. For example, the I/O controller hub **1430** may be integrated within the one or more processor **1402**, or the memory controller hub **1416** and I/O controller hub **1430** may be integrated into a discreet external graphics processor, such as the external graphics processor **1412**.

(148) Having described several embodiments of the techniques in detail, various modifications and improvements will readily occur to those skilled in the art. Such modifications and improvements are intended to be within the spirit and scope of the invention. Accordingly, the foregoing description is by way of example only, and is not intended as limiting. For example, any components described above (e.g., FIGS. 3-6C, 8-10 and 13-14) may comprise hardware, software or a combination of hardware and software. In a non-limiting example, voice isolation network as described above, or a portion of the voice isolation network, may be implemented in software by a processor of the ear-worn device. Alternatively, the voice isolation network may be implemented in hardware, e.g., in an FPGA or an ASIC chip installed in the ear-worn device.

(149) FIG. 15 illustrate an example circuit including a voice isolation network and a voice signature network coupled to the voice isolation work, where the circuit (hardware-implemented or software-implemented) may be implemented in an ear-worn device, or in an external electronic device. For example, circuitry **1500** may be implemented in an ear-worn device **108** (in FIGS. 1-2), **302** (in FIG. 3), or implemented as an expanded voice isolation network **403** (in FIG. 4A), **450** (in FIG. 4B), **540** (in FIG. 5A), **570** and **580** (in FIG. 5B), or **600** (in FIG. 6A). In some embodiments, circuitry **1500** may include a combination of configurations spread between an ear-worn device and an electronic device, as described above in connection with other components of an audio system including an ear-worn device and an electronic device. For example, circuitry **1500** may include both a voice isolation network **1502** configured to isolate the speech of speakers from an audio signal, and a voice signature network **1504** configured to provide the voice signature(s) of target speaker(s) to the voice isolation network **1502** from an audio signal. Voice isolation network **1502** may be configured in a similar manner as voice isolation network **600** of FIG. 6A. Voice signature network **1504** may be configured in a similar manner as voice signature network **1002** of FIG. 10, with the difference being that the circuitry **1000** (or a portion of circuitry **1000** including voice signature network **1002**) may be implemented in the ear-worn device. In such configuration, the input audio signal **1514** to the voice signature network **1504** may be provided directly by the ear-worn device and the output of the voice signature network may be provided directly to the voice isolation network **1502**. The output audio signal from the voice isolation network **1502** may be provided to other components in the ear-worn device for further process as previously described

and output to the receiver(s) of the ear-worn device.

(150) In another variation, one or more components in an ear-worn device described above may be implemented in an electronic device. For example, circuit **1500** may be implemented in an external electronic device such as **110** (in FIGS. **1-2**), **304** (in FIG. **3**). In such case, the input audio signal **1512** to the voice isolation network **1502** may be transmitted from the ear-worn device or detected directly by the microphone of the electronic device. The output audio signal **1518** of the voice isolation network may be transmitted to the ear-worn device for further processing as described above, and output to the receiver(s) of the ear-worn device.

(151) In other variations, the voice signatures may be extracted and/or stored on any suitable device. For example, the voice signatures of speakers or a subset of speakers (e.g., known speakers, or default target speakers) may be stored on the ear-worn device itself. Alternatively, and/or additionally, voice signatures of some or all speakers may be extracted using an on-board voice signature network on the ear-worn device, instead of on an external electronic device. In other variations, the voice signature may be extracted without using a voice signature network. For example, a voice signature of a speaker may include a feature vector that contains the average speech power at different frequencies of an audio signal including the speech of the speaker. This voice signature may be obtained using a traditional signal processing technique. Similarly, ear-worn device may have various user interfaces that allow a wearer to make some user selections as described in connection with FIGS. **11A-11F**. For example, an ear-worn device may have a user interface including one or more buttons, one or more lights, an audio interface, and/or a visual interface, which allows a user to make a variety of user selections in connection with one or more operations of the ear-worn device, and/or one or more operations performed on the external electronic device described in various embodiments above.

(152) In further variations, the voice isolation network may be adapted to handle scenarios when multiple target speakers speak contemporaneously in a conversation, or when multiple target speakers and other non-target speaker(s) speak contemporaneously. In such cases, the voice isolation network training system may re-arrange the training dataset or re-create the training dataset to have clips that are mixed synthetically to have multiple target speakers. The training system may train the machine learning model with various permutations of multiple target speakers and multiple non-target speakers in a given clip. Other methods may also be possible.

(153) As should be appreciated from the foregoing, according to some embodiments of the technology described herein, an ear-worn device is configured to use a voice isolation machine learning model that operates on a voice signature to isolate a component of a received speech signal representing speech of a target speaker or non-target speaker associated with the voice signature. The voice signature may be obtained from a separate machine learning model, such as a voice signature machine learning model, configured to extract the voice signature from reference audio clips representing speech of the target or non-target speaker. For example, FIG. **15** shows that a voice signature network **1504** is configured to process an audio signal (e.g., audio signal **2**) to extract voice signature(s) of target speaker(s) **1516**. The voice isolation network **1502** processes an audio signal (e.g., audio signal **1**), in conjunction with the voice signature(s) output by the voice signature network **1504**, to generate an output audio signal representing isolated speech of the target speaker(s). Such an embodiment may be advantageous for multiple reasons. As a non-limiting example, the voice signature network may be implemented on an electronic device different from the ear-worn device, thereby reducing the computational complexity of implementing multiple machine learning networks on the ear-worn device, while retaining voice isolation capabilities. In addition, multiple voice signatures may be determined by the voice signature network and supplied to the voice isolation network to allow separation of speech associated with multiple contemporaneous speakers.

(154) In further variations, a single network, such as a voice isolation and classification network, may be configured to both (a) de-noise the input audio signal and (b) determine an embedding of

the input audio signal. The embedding may be compared to a reference embedding representing a voice signature of a target or non-target speaker. The reference embedding may be generated by passing a clip of speech from a target speaker or non-target speaker through the same network and averaging the resultant embeddings. The result of the comparison may be used to classify the input audio signal as belonging to the target or non-target speaker, or as not belonging to any such target or non-target speaker. In some embodiments, the isolated component(s) of the audio signal are selectively processed based on the result of the classification. For example, an isolated component of the audio signal that is classified as originating from a target speaker may be amplified and/or enhanced, while an isolated component of the audio signal that is classified as originating from a non-target speaker may be suppressed.

(155) In some embodiments, a voice isolation and classification network may be used instead of, or in addition to, a voice isolation network separate from a voice signature network. Such an embodiment may be advantageous for multiple reasons. For example, instead of providing a voice signature as input to a machine learning model and using the machine learning model to identify and separate out speech associated with the voice signature, the voice isolation and classification network may be trained to perform the two functions described above, namely (a) de-noising the input audio signal and (b) determining an embedding of the input audio signal. Determination of whether the input audio signal includes speech of a target or non-target speaker may then be accomplished using a comparator, such as a cosine similarity comparator. Such operation may be simpler than using a machine learning model to identify speech associated with a target speaker by having the machine learning model apply a voice signature as an input, and may lead to better overall performance of the hearing system. Also, using a voice isolation and classification machine learning model that performs the functions (a) and (b) described above may allow for use of the same model on both the ear-worn device (e.g., hearing aid) and the separate electronic device (e.g., mobile phone), which may simplify training and deployment of the model compared to an embodiment in which separate machine learning models are used for voice signature detection and voice isolation. Moreover, an audio clip representing the voice of a speaker may be passed through the voice isolation and classification model once, and then used to predict whether subsequent input audio signals represent speech from the same speaker. As another non-limiting example, layers of the voice isolation and classification machine learning model can take advantage of information already processed in previous frames. In other words, the layers that generate the discriminative embedding can also be recurrent, so while processing audio frame-by-frame in real-time, the network can use recently received information to identify who is speaking in a given frame (which is done by generating an embedding for a given frame).

(156) FIG. 16 illustrates an example of a voice isolation network having a voice isolation model for de-noising an input audio signal and determining an embedding of the input audio signal. The illustrated voice isolation network **1600** includes a voice isolation model **1602** configured to receive an audio signal **1601**. The voice isolation model **1602** includes a voice isolation component **1604** and an embedding component **1606**. Those two components may be considered separate components of the same machine learning model, namely voice isolation model **1602**. The audio signal **1601** is first processed by the voice isolation component **1604** to produce a de-noised audio signal **1605**. The de-noised audio signal is provided to the embedding component **1606** which determines an embedding **1610** of the audio signal. The voice isolation model **1602** therefore may provide two outputs, including the de-noised audio signal **1605** and the embedding **1610**.

(157) The illustrated voice isolation network may be used to generate a voice signature of a target or non-target speaker, represented by the embedding **1610**. The audio signal **1601** may be an audio clip of speech of the target or non-target speaker. Therefore, the embedding may be a discriminative embedding representing a voice signature of the target or non-target speaker. That discriminative embedding may then be used as a reference embedding for later comparison to determine whether received audio includes speech from the target or non-target speaker. In some

embodiments, the audio clip of the speech will be processed through the network frame by frame, and then the embedding averaged across all the frames will be the discriminative embedding that is stored for later comparison. In some embodiments, the non-target speaker is the wearer of the ear-worn device, and the embedding **1610** may represent a voice signature of the wearer of the ear-worn device.

(158) FIG. **17** illustrates a voice isolation and classification network according to some embodiments of the present technology. The illustrated voice isolation and classification network **1700** includes a voice isolation model **1702** having a voice isolation component **1704** and an embedding component **1706**. In some embodiments, the voice isolation model **1702** is the same model as voice isolation model **1602**, but the two may be different instances of the same machine learning model in different locations. For instance, the voice isolation model **1602** may be used on an electronic device (e.g., a mobile phone) and the voice isolation model **1702** may be used on an ear-worn device (e.g., a hearing aid).

(159) The voice isolation model **1702** receives an audio signal **1701** as input and generates a de-noised audio signal **1705** using the voice isolation component **1704**. The voice isolation model **1702** further determines an embedding **1710** using the embedding component **1706**. The voice isolation model **1702** therefore may output the de-noised audio signal **1705** and the embedding **1710**.

(160) The voice isolation and classification network **1700** further comprises a comparator **1712**. The comparator **1712** is configured to compare the embedding **1710** with a reference embedding **1714**. The reference embedding may be provided by a separate instance of the voice isolation model, for example by voice isolation model **1602**. In some embodiments, then, the voice isolation model **1602** may be used to generate an embedding **1610** which may be stored and used as a reference embedding for use by a different instance of the voice isolation model. In one embodiment, for example, the voice isolation model **1602** is used on a mobile phone to determine a voice signature of a target or non-target speaker represented by embedding **1610**. The embedding **1610** is then provided to an ear-worn device having the voice isolation and classification network **1700**, to be used as the reference embedding **1714**. The embedding **1610** in some embodiments represents a voice signature of the wearer of the ear-worn device.

(161) The comparator **1712** compares the embedding **1710** with the reference embedding **1714** (e.g., embedding **1610** from FIG. **16**) and determines whether they match. If the embedding **1710** matches the reference embedding **1714**, the comparator **1712** outputs a classification value **1716** (e.g., a 1) indicating the match and thus that the audio signal **1701** includes speech from a speaker associated with reference embedding **1714**. In some embodiments, the match may be determined by the cosine similarity between the embedding vectors, which may involve calculating the cosine similarity between the embedding vectors. In other embodiments, this may be another small recurrent network that predicts the degree of match. If the embedding **1710** does not match the reference embedding **1714**, the comparator **1712** outputs a classification value **1716** (e.g., a 0) indicating no match and thus that the audio signal **1701** does not include speech from the speaker associated with the reference embedding **1714**.

(162) The classification value **1716** is provided to a relative gain filter **1718** which processes the de-noised audio signal **1705**. For example, if the classification value **1716** indicates a match between the embedding **1710** and the reference embedding **1714**, output audio signal **1720** may be enhanced or attenuated according to user preferences for the speaker associated with the reference embedding **1714**. For example, if the reference embedding **1714** represents the wearer's own voice signature, indication of a match between the embedding **1710** and the reference embedding **1714** may result in the relative gain filter **1718** outputting an attenuated output audio signal **1720**. In this manner, the wearer's own voice may be suppressed and the wearer may have a more positive experience with the hearing aid.

(163) The output audio signal **1720** may be provided to a DSP (e.g., see FIGS. **4A** and **4B**) for

outputting to the receiver(s) of the ear-worn device.

(164) The comparator **1712** may be considered part of or separate from the voice isolation model. In the representation of FIG. **17**, the comparator **1712** is shown as separate from the voice isolation model **1702**. However, in alternative embodiments, the comparator **1712** may be considered to form part of the voice isolation model.

(165) According to an embodiment of the present application, the voice isolation model **1602** and the voice isolation and classification network **1700** are used together. The voice isolation model **1602** is used on an electronic device (e.g., electronic device **110**) to generate an embedding **1610** representing a target or non-target speaker. The de-noised audio signal **1605** may not be used. The voice isolation and classification network **1700** is used on an ear-worn device (e.g., ear-worn device **108**, which may be a hearing aid) and receives the embedding **1610** from the electronic device for use as reference embedding **1714**. Thus, it should be appreciated that in some embodiments the same voice isolation model may be used on both the electronic device and the ear-worn device. This system architecture simplifies development and training of the machine learning model compared to a hearing system that uses different machine learning models on the electronic device and ear-worn device.

(166) Training of the voice isolation model **1602** and voice isolation model **1702**—which, again, may be different instances of the same machine learning model may be done in stages. First, the model may be trained to do the task of voice isolation. After voice isolation layers are trained, then additional layers are trained to check whether the isolated voice is a match for a given snippet of voice audio. The trained model would therefore output both clean speech and the result of a classifier that matched the speech to a voice signature. The trained model may then be used to amplify or suppress the voice. For example, when the target voice is present, the voice stream can be played out at full volume. When the classifier determines that a different speaker is speaking, the voice stream can be suppressed.

(167) As described above, such an approach may provide various benefits. For example, the voice signature clip the represents the target or non-target voice can be passed through the voice isolation network just once, and then stored as a single vector representing the average embedding for the entire voice signature clip. The subsequently received input audio signal may be processed frame by frame with low latency (e.g., in real time), and the voice isolation and classification network can predict the discriminative embedding. Also, since the machine learning model may be a recurrent network, the predicted embedding for a given frame of the input audio signal can utilize information from previous frames.

(168) The voice isolation networks of FIGS. **16** and **17** may also be used with multiple target or non-target speakers. To do so, just the classifier layers of the voice isolation and classification model are rerun for each target or non-target speaker. The layers of the machine learning model that perform de-noising may be run once for each audio frame. Since the layers that perform de-noising may be computationally expensive, such operation may allow for relatively computationally little effort to process input audio signals to identify multiple target or non-target speakers.

(169) The various embodiments described in FIGS. **1-17** provide advantages over conventional hearing aids. For example, rather than separating out speech from noise as done in conventional hearing aids (in which all voices may be presented equally to the wearer), techniques described herein allow a system to selectively isolate one or more target speakers from other non-target speakers or noise, thus provide a positive experience for the wearer of the ear-worn device. Such positive experience allows the wearer to focus on the speech of a subset of the voices in the environment, which is a function important to natural hearing.

(170) Further, the use of voice signatures in a voice isolation network allows the system to selectively isolate the target speakers. Other techniques, such as the use of triplets of clips combined with the use of a trained voice signature machine learning model in a training system yields a trained voice isolation machine learning model with improved performance on isolating

speech(es) of target speaker(s).

(171) Various techniques are provided to further process the isolated speech by preferentially treating the isolated speech, to generate enhanced speech for target speakers with increased SNR. Other techniques include dynamically controlling the voice isolation network to activate or deactivate during a conversation. Such controlling results in a saving of computation, which makes it possible to execute a machine learning model in real-time and on a power-restraint ear-worn device.

(172) Even further, the system uses a phone associated with the wearer of the ear-worn device, which allows the wearer to effectively select target speakers to whom the wearer of the ear-worn device prefers to listen. For example, the phone may store a registry of known speakers, which may be updated by the user. The registry of known speakers thus assists the user to quickly select target speakers in a multi-speaker conversation.

(173) According to some embodiments, a method of operating a mobile processing device operatively couplable to an ear-worn device is provided. The method comprises: wirelessly transmitting, from the mobile processing device to the ear-worn device, a voice signature of at least one target speaker.

(174) According to some embodiments, an apparatus is provided. The apparatus comprises at least one processor; and at least one non-transitory computer-readable medium storing instructions that, when executed, cause the at least one processor to perform a method of operating a mobile processing device operatively couplable to an ear-worn device. The method comprises: wirelessly transmitting, from the mobile processing device to the ear-worn device, a voice signature of at least one target speaker.

(175) According to some embodiments, a non-transitory computer-readable medium comprising instructions that, when executed, cause at least one processor to perform a method of operating a mobile processing device operatively couplable to an ear-worn device. The method comprises: wirelessly transmitting, from the mobile processing device to the ear-worn device, a voice signature of at least one target speaker.

(176) In some embodiments, the voice signature comprises a feature vector, and wirelessly transmitting the voice signature of the target speaker comprises wirelessly transmitting the feature vector.

(177) Some embodiments further comprise storing a registry of plurality of speakers including the target speaker on the mobile processing device, wherein the registry comprises a plurality of entries respectively each associated with a voice signature of a respective speaker of the plurality of speakers; wherein transmitting the voice signature of the target speaker comprises transmitting an identifier identifying the voice signature of the target speaker in the registry.

(178) Some embodiments further comprise receiving an input speech signal including speech from the target speaker; and determining that the input speech signal includes the speech from the target speaker by processing the input speech signal with a machine learning model that isolates speech associated with the voice signature of the target speaker.

(179) In some embodiments, receiving the input speech signal comprises receiving the input speech signal from a microphone coupled to the mobile processing device or from the ear-worn device.

(180) Some embodiments further comprise determining that the input speech signal includes speech from an additional speaker besides the target speaker; and wirelessly transmitting from the mobile processing device to the ear-worn device, a voice signature of the additional speaker.

(181) Some embodiments further comprise receiving a user selection identifying the target speaker and/or the additional speaker, and wherein wirelessly transmitting the voice signature of the target speaker and/or wirelessly transmitting the voice signature of the additional speaker is performed in response to receiving the user selection.

(182) Some embodiments further comprise, before receiving the user selection identifying the target speaker and/or the additional speaker, displaying respective entries of the target speaker

and/or the additional speaker in the registry.

(183) Some embodiments further comprise receiving an input speech signal including speech from the target speaker; and obtaining the voice signature of the target speaker based on the input speech signal.

(184) In some embodiments, obtaining the voice signature of the target speaker comprises developing the voice signature of the target speaker by processing the input speech signal with a machine learning model.

(185) In some embodiments, the mobile processing device comprises a microphone, and wherein receiving the input speech signal comprises receiving the input speech signal from the microphone.

(186) In some embodiments, receiving the input speech signal comprises wirelessly receiving the input speech signal from the hearing aid.

(187) Some embodiments further comprise receiving a user selection identifying the target speaker, and wherein wirelessly transmitting the voice signature of the target speaker is performed in response to receiving the user selection.

(188) In some embodiments, the mobile processing device stores a plurality of voice signatures including the voice signature of the target speaker. Some embodiments further comprise, in advance of receiving the user selection identifying the target speaker, presenting the user with an option to select the target speaker.

(189) In some embodiments, each voice signature in the plurality of voice signatures is associated with a respective speaker among a plurality of speakers in a contact list including the target speaker, and storing the plurality of voice signatures comprises: collecting a respective audio segment for each of the plurality of speakers; generating a respective voice signature for each of the plurality of speakers using a machine learning model over the respective audio segment; and registering in a registry the voice signatures of the plurality of speakers with the plurality of speakers in the contact list.

(190) In some embodiments, collecting the respective audio segment for a speaker of the plurality of speakers comprises: displaying a prompt; recording the respective audio segment in response to the speaker reading the prompt.

(191) In some embodiments, collecting the respective audio segment for a speaker of the plurality of speakers comprises processing an audio signal recorded in a conversation including the speaker.

(192) According to some embodiments, a method of selectively processing, with an ear-worn device including a processor and a microphone coupled to the processor, a target speaker's speech from an audio signal comprising temporally overlapping speech components from multiple speakers is provided. In some embodiments, the target speaker comprises a wearer of the ear-worn device. The method comprises: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; isolating, with the processor of the ear-worn device, a component of the audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers by processing the audio signal with a machine learning model; and suppressing the isolated component of the audio signal representing the target speaker's speech.

(193) The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

(194) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified.

(195) As used herein in the specification and in the claims, the phrase “at least one,” in reference to

a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified.

(196) The terms “approximately” and “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, within $\pm 5\%$ of a target value in some embodiments, and yet within $\pm 2\%$ of a target value in some embodiments. The terms “approximately” and “about” may include the target value.

(197) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(198) Having described above several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be object of this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A method of selectively processing, with an ear-worn device including a processor and a microphone coupled to the processor, a target speaker's speech from an audio signal comprising temporally overlapping speech components from multiple speakers, the method comprising: detecting the audio signal with the microphone of the ear-worn device; providing the audio signal detected by the microphone of the ear-worn device to the processor of the ear-worn device; and isolating, with the processor of the ear-worn device, a component of the audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers by: providing the audio signal as a first input to a machine learning model and providing a voice signature of the target speaker as a second input to the machine learning model; and processing the audio signal with the machine learning model using the voice signature of the target speaker.
2. The method of claim 1, wherein the ear-worn device is a hearing aid or a headphone.
3. The method of claim 1, wherein the voice signature of the target speaker is a feature vector obtained from another machine learning model trained to discriminate between voices of speakers, and wherein providing the voice signature as the second input to the machine learning model comprises providing the feature vector as an input signal to the machine learning model.
4. The method of claim 1, wherein the audio signal is a second audio signal, and wherein the method further comprises: detecting a first audio signal with the microphone of the ear-worn device; and extracting the voice signature of the target speaker from the first audio signal before processing the second audio signal with the machine learning model using the voice signature of the target speaker.
5. The method of claim 1, wherein processing the audio signal with the machine learning model comprises processing the audio signal with a neural network.
6. The method of claim 5, wherein processing the audio signal with a neural network comprises processing the audio signal with a recurrent neural network or a convolutional neural network.
7. The method of claim 1, wherein the target speaker comprises a wearer of the ear-worn device, and wherein the method further comprises, after isolating the component of the audio signal representing the target speaker's speech from among the temporally overlapping speech

components, suppressing the component of the audio signal representing the target speaker's speech.

8. The method of claim 1, further comprising amplifying the component of the audio signal representing the target speaker's speech after isolating the component of the audio signal representing the target speaker's speech.

9. The method of claim 1, wherein the temporally overlapping speech components from multiple speakers comprises the component of the audio signal representing the target speaker's speech and a component of the audio signal representing a secondary speaker's speech, and wherein the method further comprises suppressing the component of the audio signal representing the secondary speaker's speech after isolating the component of the audio signal representing the target speaker's speech.

10. The method of claim 1, wherein the ear-worn device further comprises a speaker device, and wherein the method further comprises outputting from the speaker device of the ear-worn device the component of the audio signal representing the target speaker's speech after isolating the component of the audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers.

11. An ear-worn device comprising: one or more microphones configured to detect an audio signal including speech from one or more speakers in a conversation; at least one processor configured to perform the method of claim 1 to generate an output audio signal; and one or more speaker devices configured to output the output audio signal.

12. A system, comprising: the ear-worn device of claim 11; and a second ear-worn device comprising: one or more second microphones configured to detect a second audio signal; at least one second processor configured to: generate a second output audio signal at least in part by: isolating a component of the second audio signal representing the target speaker's speech from among the temporally overlapping speech components from multiple speakers by processing the second audio signal with a second machine learning model; receive, from the ear-worn device, the output audio signal; and combine the output audio signal and the second output audio signal to generate a combined audio signal; and one or more second speaker devices configured to output the combined audio signal.
